Complex Analysis

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Complex analysis is the study of differentiable functions of a complex variable. Asking for a function $\mathbb{C} \to \mathbb{C}$ to be differentiable turns out to be a shockingly strong requirement – such functions are automatically infinitely differentiable, are determined on huge regions by their values on tiny ones, and their integrals along closed curves detect delicate information about singularities. The theory that develops from this single definition is remarkably rigid, remarkably beautiful, and (as we will see towards the end of these notes) genuinely useful for down-to-earth tasks like computing real integrals.

This article constitutes my notes for the 'Complex Analysis' course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 Basic Notions

§1.1 Complex Differentiation

This course is about complex valued functions of a (single) complex variable, that is, functions

$$f : A \to \mathbb{C}, \quad \text{where } A \subseteq \mathbb{C}.$$

In this course we will focus on differentiable functions f (the definition of which we will give soon), but let's start by recalling the notion of continuity.

Definition 1.1 (Continuity)

A function $f : A \to \mathbb{C}$ is **continuous** at a point $w \in A$ if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that if $z \in A$ with $|z - w| < \delta$ then $|f(z) - f(w)| < \varepsilon$.

Note that by identifying $\mathbb{C}$ with $\mathbb{R}^2$ in the usual way, we can write $f(z) = u(x, y) + iv(x, y)$ for $z = x + iy \in A$ with $u, v : A \to \mathbb{R}$. With this, the triangle inequality implies that f is continuous at $w = c + id$ if and only if u, v are continuous at (c, d).

We can now define the notion of differentiation for complex functions.

Definition 1.2 (Complex Differentiation)

Let $f : U \to \mathbb{C}$ where $U \subseteq \mathbb{C}$ is open. We say that f is **differentiable** at $w \in U$ if the limit

$$f'(w) = \lim_{z \to w} \frac{f(z) - f(w)}{z - w}$$

exists as a complex number.

Definition 1.3 (Holomorphic)

We say that f is **holomorphic at $w \in U$** if there is $\varepsilon > 0$ such that $D(w, \varepsilon) \subset U$ where f is differentiable at every point in $D(w, \varepsilon)$. We say that f is **holomorphic** if it is holomorphic at every point in U.

Remark. Sometime authors use the word 'analytic' instead of the word 'holomorphic'.

Definition 1.4 (Entire)

If $f : \mathbb{C} \to \mathbb{C}$ is holomorphic on all of $\mathbb{C}$, we say f is **entire**.

It's straightforward to check that the same rules of differentiation for real variables hold for complex functions (product rule, quotient rule, chain rule, etc), which makes computing complex derivatives relatively straightforward.

Example 1.5 (Complex Differentiable Functions)

(i) Polynomials $p(z) = \sum_{k=0}^{n} a_k z^k$, $a_0, \dots, a_n \in \mathbb{C}$ are holomorphic on all of $\mathbb{C}$.

(ii) If p and q are polynomials, then p/q is holomorphic on $\mathbb{C} \backslash \{z \mid q(z) = 0\}$.

We saw above that the notion of continuity is basically the same for both real and complex variables. A natural question is then if the same is true for complex differentiability. Indeed, the answer is no! Complex differentiability is a much stronger condition than asserting real differentiability on the real and imaginary components of a function.

We do however have a criterion for establishing complex differentiability from the real and imaginary parts.

Theorem 1.6 (Cauchy-Riemann Equations)

A function $f = u + iv : U \to \mathbb{C}$ is differentiable at $w = c + id \in U$ if and only if $u, v : U \to \mathbb{R}$ are differentiable at $(c, d) \in U$ *and* u, v satisfy the **Cauchy-Riemann** equations at (c, d):

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \quad \text{at } (c, d). \end{aligned}$$

If f is differentiable at $w = c + id$, then $f'(w) = \frac{\partial u}{\partial x}(c, d) + i \frac{\partial v}{\partial x}(c, d)$.

Proof. The function f is differentiable at w with derivative $f'(w) = p + iq$ if and only if

$$\lim_{z\to w}\frac{f(z)-f(w)}{z-w}=p+iq \iff \lim_{z\to w}\frac{f(z)-f(w)-(z-w)(p+iq)}{|z-w|}=0.$$

Separating real and imaginary parts and writing $f=u+iv$, the above holds if and only if

$$\lim_{(x,y)\to(c,d)}\frac{u(x,y)-u(c,d)-p(x-c)+q(y-d)}{\sqrt{(x-c)^2+(y-d)^2}}=0$$

and

$$\lim_{(x,y)\to(c,d)}\frac{v(x,y)-v(c,d)-q(x-c)-p(y-d)}{\sqrt{(x-c)^2+(y-d)^2}}=0.$$

That is, if and only if u is differentiable at (c,d) with $Du(c,d)(x,y)=px-qy$, and v is differentiable at (c,d) with $Dv(c,d)(x,y)=qx+py$.

Taking partial derivatives, this is true if and only if u,v are differentiable at (c,d) and $u_x(c,d)=p=v_y(c,d)$ and $u_y(c,d)=-q=-v_x(c,d)$, that is, the Cauchy-Riemann equations hold at (c,d).

We also get from the above that if f is differentiable at w, then $f'(w)=p+iq=u_x(c,d)+iv_x(c,d)$. □

Remark (Warning)**.** The real and imaginary parts u,v satisfying the Cauchy-Riemann equations at a point does *not* guarantee the differentiability of f, and a counterexample is given on the first example sheet.

We can easily use this criterion to show that many functions are not complex differentiable.

Example 1.7 (Conjugation isn't Differentiable)

Consider the function $f(z)=\overline{z}=x-iy$. For this, $u(x,y)=x$ and $v(x,y)=-y$, so $u_x=1$ and $u_y=0$, $v_x=0$ and $v_y=-1$. So the Cauchy-Riemann equations do not hold anywhere, and f is not differentiable at any point.

Now, if we just wanted to show that the differentiability of $f=u+iv$ at $w=c+id$ implies that the partial derivatives u_x,u_y,v_x and v_y exist at (c,d) and satisfy the Cauchy-Riemann equations, then we can proceed more simply as follows.

Begin with the definition of differentiability

$$f'(w)=\lim_{h\to 0}\frac{f(w+h)-f(w)}{h}.$$

Then taking $h=t\in\mathbb{R}$, we get

$$f'(w)=\lim_{t\to 0}\left(\frac{u(c+t,d)-u(c,d)}{t}+i\frac{v(c+t,d)-v(c,d)}{t}\right),$$

from which the real and imaginary parts imply that the limits

$$\lim_{t\to 0}\frac{u(c+t,d)-u(c,d)}{t}\quad\text{and}\quad\lim_{t\to 0}\frac{v(c+t,d)-v(c,d)}{t}$$

both exist, that is, $u_x(c,d)$ and $v_x(c,d)$ exists, and $f'(w)=u_x(c,d)+iv_x(c,d)$. Similarly, taking $h=it$ with $t\in\mathbb{R}$ gives $f'(w)=v_y(c,d)-iu_y(c,d)$, so $u_x=v_y$ and $u_y=-v_x$ at (c,d).

Because partial derivatives are a little easier to work with than the standard derivative of a function on $\mathbb{R}^2$, we can use a straightforward corollary of the Cauchy-Riemann equations for checking complex differentiability[1].

Corollary 1.8 (Cauchy-Riemann and Continuous Partials implies Holomorphic)

Let $f = u + iv : U \to \mathbb{C}$. If u, v have continuous partial derivatives at a point $(c, d) \in U$ and satisfy the Cauchy-Riemann equations at (c, d), then f is differentiable at $w = c + id$.

Proof. Continuity of partial derivatives of u implies that u is differentiable, and similarly for v. So this follows from our original theorem. □

Complex differentiability is a *much* stronger notion than the differentiability of real and imagine parts. This leads to some surprising theorems compared to the real case. For example:

(a) *Liouville's Theorem.* If $f : \mathbb{C} \to \mathbb{C}$ is holomorphic and bounded then f is constant.

(b) If $f : U \to \mathbb{C}$ is holomorphic, then f is automatically infinitely differentiable on U.

We will prove these and much more later on.

Remark (Harmonic Functions)**.** Note that the second theorem implies that the partial derivatives of u, v exist to all orders, so we can differentiate the Cauchy-Riemann equations to get

$$u_{xx} = v_{yx}, \quad \text{and} \quad u_{yy} = -v_{xy},$$

and since $v_{yx} = v_{xy}$, we get

$$\nabla^2 u = u_{xx} + u_{yy} = 0 \quad \text{in } U,$$

and similarly $\nabla^2 v = 0$ in U. So *real and imaginary parts of a holomorphic function are harmonic.*

We will now consider curves in the complex plane.

Definition 1.9 (Curve)

A **curve** is a continuous map $\gamma : [a, b] \to \mathbb{C}$, where $[a, b] \subset \mathbb{R}$ is a closed interval. We say γ is a **C^1 curve** if γ' exists and is continuous on $[a, b]$.

Definition 1.10 (Path Connected)

An open set $U \subset \mathbb{C}$ is **path connected** if for any two points $z, w \in U$ there is a curve $\gamma : [0, 1] \to U$ such that $\gamma(0) = z$ and $\gamma(1) = w$.

Definition 1.11 (Domain)

A **domain** is a non-empty, open, path connected subset of $\mathbb{C}$.

[1] Remarkably, the *Looman-Menchoff* theorem says that it suffices for u and v to be continuous and satisfy the Cauchy-Riemann equations to achieve complex differentiability, but this is quite non-trivial to prove.

Corollary 1.12 (Zero Derivative implies Constant)

If $U \subset \mathbb{C}$ is a domain and $f : U \to \mathbb{C}$ is holomorphic with $f'(z) = 0$ for every $z \in U$, then f is constant.

Proof. Write $f = u + iv$. By the Cauchy-Riemann equations, $f' = 0$ implies that $Du = 0$ and $Dv = 0$ in U. Then since U is a domain, we get that u and v are constant. □

§1.2 Power Series

So far we've only seen very few explicit holomorphic functions (namely polynomials on $\mathbb{C}$ and rational functions on their domains). We'd like to generate more, which we can do by looking at *power series*.

Recall the notion of *radius of convergence*.

Theorem 1.13 (Radius of Convergence)

If $c_n \in \mathbb{C}$ is a sequence of complex numbers, then there is a unique $R \in [0, \infty]$, the **radius of convergence**, such that the power series

$$\sum_{n=0}^{\infty} c_n (z-a)^n, \quad z, a \in \mathbb{C}$$

converges absolutely if $|z - a| < R$ and diverges if $|z - a| > R$. If $0 < r < R$, then the series converges uniformly (with respect to z) on the compact disk $\overline{D(a, r)} = \{z \in \mathbb{C} : |z - a| \leq r\}$.

Note that there are no claims about convergence when $|z - a| = R$ for $R > 0$. There are also various expressions for R, for example

$$R = \sup\{r \geq 0 \mid |c_n| r^n \to 0 \text{ as } n \to \infty\}$$

and

$$\frac{1}{R} = \limsup_{n \to \infty} |c_n|^{1/n}.$$

We get quite natural behavior of power series with respect to complex differentiability.

Theorem 1.14 (Derivatives of Power Series)

Let $\sum_{n=0}^{\infty} c_n (z-a)^n$ be a power series with radius of convergence $R > 0$. Fix some $a \in \mathbb{C}$, and define $f : D(a, R) \to \mathbb{C}$ by $f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$. Then

(i) f is holomorphic on $D(a, R)$;

(ii) the derived series $\sum_{n=1}^{\infty} n c_n (z-a)^{n-1}$ also has radius convergence R, and is equal to $f'(z)$ for $z \in D(a, R)$.

(iii) f has derivatives of all orders on $D(a, R)$, and $c_n = \frac{f^{(n)}(a)}{n!}$.

(iv) if f vanishes on $D(a, \varepsilon)$ for some $\varepsilon > 0$, then $f \equiv 0$ on $D(a, R)$.

Proof. Parts (i) and (ii). Without loss of generality, take $a = 0$. Now we have $f(z) = \sum_{n=0}^{\infty} c_n z^n$ for $z \in D(0, R)$, where R is the radius of convergence.

The derived series $\sum_{n=1}^{\infty} n c_n z^{n-1}$ will have some radius of convergence $R_1 \in [0, \infty]$. To see that $R_1 \geq R$, let $z \in D(0, R)$, and choose ρ such that $|z| < \rho < R$. Then

$$n|c_n||z|^{n-1} = \left(n \left| \frac{z}{\rho} \right|^{n-1} \right) |c_n|\rho^{n-1} \leq |c_n|\rho^{n-1}$$

for sufficiently large n (as $n|z/\rho|^{n-1} \to 0$ as $n \to \infty$). Since $\sum |c_n|\rho^n$ converges, it follows that $\sum_{n=1}^{\infty} n|c_n||z|^{n-1}$ converges. Thus $D(0, R) \subset D(0, R_1)$ that is, $R_1 \geq R$.

Also, since $|c_n||z|^n \leq n|c_n||z|^n = |z|(n|c_n||z|^{n-1})$, if $\sum n|c_n||z|^{n-1}$ converges then so does $\sum |c_n||z|^n$, so $R_1 \leq R$. This implies that we must have $R_1 = R$.

To prove that f is differentiable with $f'(z) = \sum_{n=1}^{\infty} n c_n z^{n-1}$, fix some $z \in D(0, R)$. Then our key idea is that this assertion is equivalent to the continuity at z of the function $g : D(0, R) \to \mathbb{C}$ with

$$g(w) = \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } w \neq z, \\ \sum_{n=1}^{\infty} n c_n z^{n-1} & \text{if } w = z. \end{cases}$$

By substituting for f, we get that $g(w) = \sum_{n=1}^{\infty} h_n(w)$ for $w \in D(0, R)$ where h_n is a sequence of functions given by

$$h_n(w) = \begin{cases} \frac{c_n(w^n - z^n)}{w - z} & \text{if } w \neq z, \\ n c_n z^{n-1} & \text{if } w = z. \end{cases}$$

Now h_n is continuous on $D(0, R)$ for all n, then using that $\frac{w^n - z^n}{w - z} = \sum_{j=0}^{n-1} z^j w^{n-1-j}$, we get that for any r with $|z| < r < R$ and any $w \in D(0, r)$ we have $|h_n(w)| \leq n|c_n|r^{n-1}$. Defining $M_n = n|c_n|r^{n-1}$, we can note that $\sum M_n < \infty$, so by the Weierstrass M-test, $\sum h_n$ converges uniformly on $D(0, r)$. But a uniform limit of continuous functions is continuous, so $g = \sum h_n$ is continuous in $D(0, r)$ and in particular at z, as required.

Part (iii). Repeatedly apply (ii), and the formula $c_n = f^{(n)}(a)/n!$ follows by differentiating the series n times and setting $z = a$.

Part (iv). If $f = 0$ on $D(a, \varepsilon)$, then $f^{(n)}(a) = 0$ for all n, so $c_n = 0$ for all n. Hence $f = 0$ on $D(a, R)$. □

Remark. This theorem provides a way to generate a large class of holomorphic functions on a disk. Later we will show that every holomorphic function is *locally* given by a power series. Once we have that, (iii) above gives that holomorphic functions are automatically infinitely differentiable in their domain.

§1.3 The Exponential and the Logarithm

Let's now consider a specific example of a power series, the *complex exponential* function, which you are likely familiar with.

Definition 1.15 (Complex Exponential)

The **complex exponential** function is defined by the power series

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Some basic properties of this function will also be familiar.

Proposition 1.16 (Properties of the Exponential)

(i) e^z is entire, with $(e^z)' = e^z$.

(ii) $e^z \neq 0$ and $e^{z+w} = e^z \cdot e^w$ for all $z, w \in \mathbb{C}$.

(iii) $e^{x+iy} = e^x(\cos y + i \sin y)$ for $x, y \in \mathbb{R}$.

(iv) $e^z = 1$ if and only if $z = 2\pi n i$ for some integer n.

(v) Let $z \in \mathbb{C}$. Then there exists $w \in \mathbb{C}$ such that $e^w = z$ if and only if $z \neq 0$.

Proof. Part (i). The radius of convergence of the series is ∞, and to see that $(e^z)' = e^z$, differentiate the series term-by-term.

Part (ii). Fix any $w \in \mathbb{C}$ and set $F(z) = e^{z+w} \cdot e^{-z}$. Then $F'(z) = -e^{z+w} \cdot e^{-z} + e^{z+w} \cdot e^{-z} = 0$, so $F(z)$ is constant, and equal to e^w. Thus $e^{z+w} \cdot e^{-z} = e^w$ for all $z, w \in \mathbb{C}$, and the result follows noting that $e^0 = 1$.

Part (iii). By the previous part, $e^{x+iy} = e^x \cdot e^{iy}$. Now use the definition of e^{iy} and the series for $\sin y$ and $\cos y$ with $y \in \mathbb{R}$.

Part (iv) and (v). Both follow directly from (iii). $\square$

We can now consider the 'inverse' of this function, the *logarithm*.

Definition 1.17 (Logarithm)

Given $z \in \mathbb{C}$, we say that $w \in \mathbb{C}$ is a **logarithm** of z if $e^w = z$.

By our above theorem, we note that z has a logarithm if and only if $z \neq 0$, and also if z *does* have a logarithm then there are infinitely many such logarithms, each differing from each other by $2\pi n i$ for some integer n.

If w is a logarithm of z, then $e^{\operatorname{Re}(w)} = |z|$, so $\operatorname{Re}(w) = \ln|z|$ (the real logarithm of the positive number $|z|$). In particular, $\operatorname{Re}(w)$ is uniquely determined by z.

Definition 1.18 (Branches of the Logarithm)

Let $U \subset \mathbb{C}\backslash\{0\}$ be open. Then a **branch** of the logarithm on U is a continuous function $\lambda : U \to \mathbb{C}$ such that $e^{\lambda(z)} = z$ for each $z \in U$.

Remark. If λ is a branch of log on U, then λ is automatically holomorphic in U, with $\lambda'(z) = 1/z$, with the proof following by direct computation of limits.

Definition 1.19 (Principal Branch of the Logarithm)
The **principal branch** of the logarithm is the function

$$\mathrm{Log} : U_1 = \mathbb{C}\backslash\{x \in \mathbb{R} : x \le 0\} \to \mathbb{C}$$

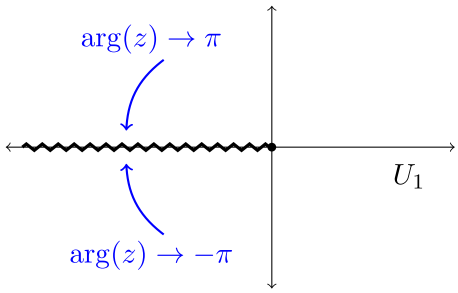
defined by

$$\mathrm{Log}(z) = \ln|z| + i\arg(z),$$

where $\arg(z)$ is the unique argument of $z \in U_1$ in $(-\pi, \pi)$.

Remark. Log does not have a continuous extension to $\mathbb{C}\backslash\{0\}$, since $\arg(z) \to \pi$ as $z \to -1$ with $\mathrm{Im}(z) > 0$, and $\arg(z) \to -\pi$ as $z \to -1$ with $\mathrm{Im}(z) < 0$. We will later see that it is impossible to define a branch of log on $\mathbb{C}\backslash\{0\}$.

The negative real axis, which we removed from the domain of Log, is called a **branch cut**. The picture to have in mind is the following: as z crosses the cut, the argument (and hence the imaginary part of any would-be logarithm) jumps by 2π.



There was nothing special about our choice to cut along the negative real axis – removing any ray from the origin (indeed, any reasonable curve joining 0 to infinity) leaves a domain on which a branch of the logarithm can be defined. The principal branch is merely the standard convention.

Proposition 1.20 (Power Series for Log)
For $|z| < 1$, we have

$$\mathrm{Log}(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}z^n}{n}.$$

Proof. Note that the radius of convergence of the series is 1 and $|z| < 1$ implies that $1 + z \in U_1$, so both sides are defined on $|z| < 1$. Then let $F(z) = \mathrm{Log}(1+z) - \sum_{n=1}^{\infty} \frac{(-1)^{n-1}z^n}{n}$, for $|z| < 1$. Then $F'(z) = \frac{1}{1+z} - \sum_{n=1}^{\infty}(-z)^{n-1} = 0$. So $F(z)$ is constant, and $F(z) = F(0) = 0$. □

Using exp and Log we can define further useful functions.

(i) For any constant $\alpha \in \mathbb{C}$, define

$$z^\alpha = e^{\alpha \mathrm{Log}(z)}, \quad z \in U_1.$$

This is the principal branch of z^α. We have that z^α is holomorphic on U_1 with $(z^\alpha)' = \alpha z^{\alpha-1}$.

(ii) We can define

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$
$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$
$$\cosh(z) = \frac{e^{z} + e^{-z}}{2}$$
$$\sinh(z) = \frac{e^{z} - e^{-z}}{2}.$$

All of these functions are entire and their derivatives are given by the familiar expressions from real variables.

§1.4 Conformal Maps

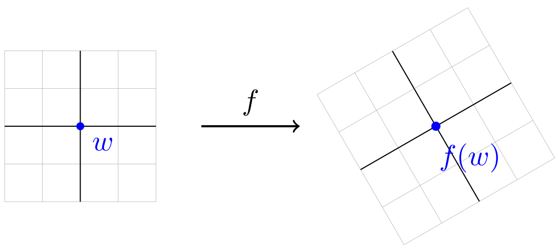
We now look at a geometric property of holomorphic functions called *conformality*. Let $f : U \to \mathbb{C}$ be holomorphic, where $U \subset \mathbb{C}$ is open, as usual. Let $w \in U$, and suppose that $f'(w) \neq 0$.

Take two C^1 curves $\gamma_1, \gamma_2 : [-1, 1] \to U$, such that $\gamma_i(0) = w$ and $\gamma_i'(0) \neq 0$. Then $f \circ \gamma_i$ are C^1 curves passing through $f(w)$. Moreover, $(f \circ \gamma_i)'(0) = f'(w)\gamma_i'(0) \neq 0$.

This implies that
$$\frac{(f \circ \gamma_1)'(0)}{(f \circ \gamma_2)'(0)} = \frac{\gamma_1'(0)}{\gamma_2'(0)},$$
and hence
$$\arg(f \circ \gamma_1)'(0) - \arg(f \circ \gamma_2)'(0) = \arg \gamma_1'(0) - \arg \gamma_2'(0).$$

This means that the angle that the curves γ_1, γ_2 make at w is the same as the angle their images $f \circ \gamma_1$ and $f \circ \gamma_2$ make at $f(w)$, in size as well as in orientation. That is, f is 'angle-preserving at w' whenever $f'(w) \neq 0$.

Geometrically, this is because multiplication by the complex number $f'(w) = re^{i\theta} \neq 0$ is a rotation by θ composed with a scaling by r. To first order, f acts near w by exactly this rotation and scaling: a small grid of curves through w is carried to a rotated and scaled grid through $f(w)$, with all angles intact.



Remark. If f is a C^1 map on U, the converse of the above also holds. That is, if $w \in U$ is such that $(f \circ \gamma)'(0) \neq 0$ for every C^1 curve γ with $\gamma(0) = w$ and $\gamma'(0) \neq 0$, and if f is angle preserving at w in the above sense, then $f'(w)$ exists and $f'(w) \neq 0$.

Definition 1.21 (Conformal)

A holomorphic function $f : U \to \mathbb{C}$ on an open set U is said to be **conformal** at the point $w \in U$ if $f'(w) \neq 0$.

Definition 1.22 (Conformal Equivalence)

Let $U, \tilde{U}$ be domains in $\mathbb{C}$. A map $f : U \to \tilde{U}$ is said to be a **conformal equivalence** between U and $\tilde{U}$ if f is a bijective holomorphic map with $f'(z) \neq 0$ for every $z \in U$.

Remark (Redundancy)**.** If f is holomorphic and injective, then $f'(z) \neq 0$ for each z (we will see this later in the course). So, in the above definition, the requirement $f'(z) \neq 0$ is redundant.

Remark (Inverse Functions)**.** It is automatic that the inverse $f^{-1} : \tilde{U} \to U$ is holomorphic. This follows from the **holomorphic inverse function theorem**, which can be proved using the real inverse function theorem.

We will now consider some examples of conformal equivalences.

Example 1.23 (Möbius Maps Give a Conformal Equivalence)

Consider the Möbius maps
$$f(z) = \frac{az + b}{cz + d}$$
where $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. This function f is conformal on $\mathbb{C} \backslash \{-d/c\}$ if $c \neq 0$, and conformal on $\mathbb{C}$ if $c = 0$.

Möbius maps sometimes serve as explicit conformal equivalences between subdomains of $\mathbb{C}$.

For example, let $\mathbb{H}$ be the open upper-half plane, $\mathbb{H} = \{z \in \mathbb{C} \mid \operatorname{Im}(z) > 0\}$. Then

$$\begin{aligned} z \in \mathbb{H} &\iff z \text{ is closer to } i \text{ than to } -i \\ &\iff |z - i| < |z + i| \\ &\iff \left| \frac{z - i}{z + i} \right| < 1. \end{aligned}$$

Thus $g = \frac{z-i}{z+i}$ maps $\mathbb{H}$ onto $D(0, 1)$, so $g : \mathbb{H} \to D(0, 1)$ is a conformal equivalence. The inverse is also a Möbius map, namely $w \mapsto i\frac{1+w}{1-w}$, so the disc and the half-plane are two interchangeable models of the 'same' domain.

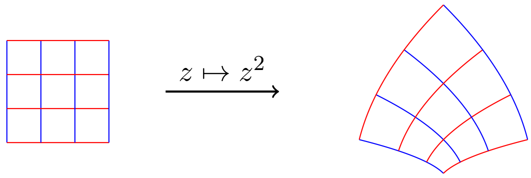
Example 1.24 (Powers Give a Conformal Equivalence)

Let $f(z) = z^n$ where $n \in \mathbb{N}$, with

$$f : \{z \in \mathbb{C} \backslash \{0\} \mid 0 < \arg(z) < \pi/n\} \to \mathbb{H}.$$

This is a conformal equivalence, with inverse $f^{-1}(z) = z^{1/n}$.

It is worth pausing to actually *look* at what a map like $z \mapsto z^2$ does. Away from the origin (where the derivative vanishes) it is conformal, so a grid of horizontal and vertical lines is carried to two families of curves (in fact parabolas) which still meet each other at right angles.



Example 1.25 (Exponentials Give a Conformal Equivalence)
The restricted exponential function given by

$$\exp : \{z \in \mathbb{C} \mid -\pi < \operatorname{Im}(z) < \pi\} \to \mathbb{C} \backslash \{x \in \mathbb{R} \mid x \leq 0\}$$

is a conformal equivalence, whose inverse is $\operatorname{Log}(z)$.

Aside: The Riemann Mapping Theorem

There is an interesting result about many domains being conformally equivalent.

Theorem 1.26 (Riemann Mapping Theorem)
Any simply connected domain $U \subset \mathbb{C}$ with $U \neq \mathbb{C}$ is conformally equivalent to $D(0, 1)$.

Here, $U \subset \mathbb{C}$ is simply connected if every continuous map $\gamma : S^1 = \partial D(0, 1) \to U$ extends to a continuous map $\Gamma : \overline{D(0, 1)} \to U$ with $\Gamma|_{\partial D(0,1)} = \gamma$. We will discuss simply connected domains in detail later on, but intuitively one should think of simply connected domains as the ones 'without holes'.

A proof of this result can be found in many textbooks on complex analysis.

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§2 Complex Integration

§2.1 Extending Riemann Integration

As we did with differentiation, we now want to extend our real (Riemann) notion of integration to the integration of complex functions $f : U \to \mathbb{C}$ with $U \subset \mathbb{C}$ along curves in U.

First we will look at complex functions of a single real variable.

Definition 2.1 (Integral on a Real Interval)
If $f : [a, b] \subset \mathbb{R} \to \mathbb{C}$ is a complex function and if f is continuous[a] then we define the **integral** of f over $[a, b]$ to be

$$\int_a^b f(t)\, \mathrm{d}t = \int_a^b \operatorname{Re}(f(t))\, \mathrm{d}t + i \int_a^b \operatorname{Im}(f(t))\, \mathrm{d}t.$$

[a] Or, more generally, f is Riemann integrable in its real and imaginary parts.

In particular,

$$\int_a^b ig(t)\ \mathrm{d}t = i\int_a^b g(t)\ \mathrm{d}t$$

for a real function $g : [a, b] \to \mathbb{R}$. Hence, by direct calculation,

$$\int_a^b wf(t)\ \mathrm{d}t = w\int_a^b f(t)\ \mathrm{d}t$$

for any $w \in \mathbb{C}$.

Proposition 2.2 (Basic Estimate)
If $f : [a, b] \to \mathbb{C}$ is continuous then

$$\left|\int_a^b f(t)dt\right| \le \int_a^b |f(t)|\ \mathrm{d}t \le (b-a)\sup_{t\in[a,b]} |f(t)|$$

with equality if and only if f is constant.

Proof. If $\int_a^b f(t)\ \mathrm{d}t = 0$ then we are done. Otherwise, write $\int_a^b f(t)\ \mathrm{d}t = re^{i\theta}$, with $\theta \in [0, 2\pi)$. Let $M = \sup_{t\in[a,b]} |f(t)|$. Then

$$\begin{aligned}\left|\int_a^b f(t)\ \mathrm{d}t\right| = r = e^{-i\theta}\int_a^b f(t)\ \mathrm{d}t &= \int_a^b e^{-i\theta} f(t)\ \mathrm{d}t \\ &= \int_a^b \mathrm{Re}(e^{-i\theta}f(t))\ \mathrm{d}t + i\int_a^b \mathrm{Im}(e^{-i\theta}f(t))\ \mathrm{d}t.\end{aligned}$$

Since the left-hand-side is real,

$$\begin{aligned}\left|\int_a^b f(t)\ \mathrm{d}t\right| &= \int_a^b \mathrm{Re}\left(e^{-i\theta}f(t)\right)\ \mathrm{d}t \\ &\le \int_a^b \left|e^{-i\theta}f(t)\right|\ \mathrm{d}t \\ &= \int_a^b |f(t)|\ \mathrm{d}t \\ &\le (b-a)M.\end{aligned}$$

Equality holds if and only if $|f(t)| = M$ and $\mathrm{Re}\left(e^{-i\theta}f(t)\right) = M$ for all $t \in [a, b]$, that is, if and only if $|f(t)| = M$ and $\arg(f(t)) = \theta$ for all $t \in [a, b]$, that is, f is constant. ☐

We can now define a more general notion of an integral along a curve in the complex plane (not just somewhere on the real line).

Definition 2.3 (Integral on a Complex Curve)
Let $U \subset \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be continuous. Let $\gamma : [a, b] \to U$ be a C^1 curve.

Then the **integral of f along γ** is

$$\int_\gamma f(z)\ \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t.$$

This integral has lots of nice basic properties that one might expect.

Proposition 2.4 (Invariance under Reparameterisation)
Let $\varphi : [a_1, b_1] \to [a, b]$ be C^1 and injective with $\varphi(a_1) = a$ and $\varphi(b_1) = b$. Let $\delta = \gamma \circ \varphi : [a_1, b_1] \to U$. Then we have

$$\int_\delta f(z)\ \mathrm{d}z = \int_\gamma f(z)\ \mathrm{d}z$$

Proof. Letting $s = \varphi(t)$, we have

$$\begin{aligned}\int_\delta f(z)\ \mathrm{d}z &= \int_{a_1}^{b_1} f(\gamma \circ \varphi(t))\gamma'(\varphi(t))\varphi'(t)\ \mathrm{d}t \\ &= \int_a^b f(\gamma(s))\gamma'(s)\ \mathrm{d}s \\ &= \int_\gamma f(z)\ \mathrm{d}z.\end{aligned}$$

The second equality follows from the definition of the integral of a function of a real variable and the change of variables formula for integrals of real functions of a real variable. □

Proposition 2.5 (Linearity)
We have

$$\int_\gamma \left(c_1 f_1(z) + c_2 f_2(z)\right)\ \mathrm{d}z = c_1 \int_\gamma f_1(z)\ \mathrm{d}z + c_2 \int_\gamma f_2(z)\ \mathrm{d}z,$$

for constants $c_1, c_2 \in \mathbb{C}$.

Proof Sketch. Check definitions. □

Proposition 2.6 (Additivity)
If $\gamma : [a, b] \to U$ is a C^1 curve and $a < c < b$, then

$$\int_\gamma f(z)\ \mathrm{d}z = \int_{\gamma|_{[a,c]}} f(z)\ \mathrm{d}z + \int_{\gamma|_{[c,b]}} f(z)\ \mathrm{d}z.$$

Proof Sketch. Check definitions. □

Proposition 2.7 (Inverse Path)

Define the inverse path $(-\gamma) : [-b, -a] \to U$ by $(-\gamma)(t) = \gamma(-t)$ for $-b \leq t \leq -a$. Then

$$\int_{(-\gamma)} f(z) \, \mathrm{d}z = -\int_{\gamma} f(z) \, \mathrm{d}z.$$

Proof Sketch. Check definitions. □

We can get the euclidean length of a curve in the complex plane by the following (which we take as a definition).

Definition 2.8 (Length of a Curve)

Let $\gamma : [a, b] \to \mathbb{C}$ be a C^1 curve. The **length** of γ is defined by

$$\operatorname{length}(\gamma) = \int_a^b \left|\gamma'(t)\right| \, \mathrm{d}t$$

Definition 2.9 (Piecewise C^1 Curve)

A **piecewise C^1 curve** is a continuous map $\gamma : [a, b] \to \mathbb{C}$ such that there exists a finite subdivision $a = a_0 < a_1 < \cdots < a_n = b$ with the property that $\gamma_j = \gamma|_{[a_{j-1}, a_j]} : [a_{j-1}, a_j] \to \mathbb{C}$ is C^1 for $1 \leq j \leq n$.

If γ is a piecewise C^1 curve as above, we *define*

$$\int_{\gamma} f(z) \, \mathrm{d}z = \sum_{j=1}^{n} \int_{\gamma_j} f(z) \, \mathrm{d}z$$

and

$$\operatorname{length}(\gamma) = \sum_{j=1}^{n} \operatorname{length}(\gamma_j) = \sum_{j=1}^{n} \int_{a_{j-1}}^{a_j} |\gamma'(t)| \, \mathrm{d}t.$$

Remark. Note that both definitions are independent of the subdivisions by the additivity property above.

With this, from now on a 'curve' will mean a piecewise C^1 curve.

Definition 2.10 (Sum of Curves)

If $\gamma_1 : [a, b] \to \mathbb{C}$ and $\gamma_2 : [c, d] \to \mathbb{C}$ are curves with $\gamma_1(b) = \gamma_2(c)$, we define the **sum** of γ_1 and γ_2 to be the curve $(\gamma_1 + \gamma_2) : [a, b + d - c] \to \mathbb{C}$, with

$$(\gamma_1 + \gamma_2)(t) = \begin{cases} \gamma_1(t) & \text{if } a \leq t \leq b, \\ \gamma_2(t - b + c) & \text{if } b \leq t \leq b + d - c. \end{cases}$$

We can then give another basic estimate for integrals over curves in the complex plane.

Proposition 2.11 (Another Basic Estimate)
For any continuous function $f : U \to \mathbb{C}$ and any curve $\gamma : [a, b] \to \mathbb{C}$, we have that

$$\left|\int_\gamma f(z)\ \mathrm{d}z\right| \le \operatorname{length}(\gamma) \sup_\gamma |f|$$

where $\sup_\gamma |f| = \sup_{t\in[a,b]} |f(\gamma(t))|$.

Proof. If γ is C^1, then

$$\begin{aligned}\left|\int_\gamma f(z)\ \mathrm{d}z\right| &= \left|\int_a^b f(\gamma(t))\gamma'(t)\ \mathrm{d}t\right| \\ &\le \int_a^b \left|f(\gamma(t))\|\gamma'(t)\right|\ \mathrm{d}t \\ &\le \sup_{t\in[a,b]} |f(\gamma(t))| \operatorname{length}(\gamma).\end{aligned}$$

If γ is piecewise C^1 then the result follows from the definition

$$\int_\gamma f(z)\ \mathrm{d}z = \sum_{j=1}^n \int_{\gamma_j} f(z)\ \mathrm{d}z$$

where γ_j is C^1, and the triangle inequality. □

§2.2 The Fundamental Theorem of Calculus

In real analysis, the fundamental theorem of calculus links differentiation and integration. The same is true for integrals along curves in the complex plane, and (as in the real case) it is the main computational tool we have for evaluating integrals directly.

Definition 2.12 (Antiderivative)
Let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be continuous. An **antiderivative** of f is a holomorphic function $F : U \to \mathbb{C}$ such that $F'(z) = f(z)$ for all $z \in U$.

Theorem 2.13 (Fundamental Theorem of Calculus)
Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$ be continuous with antiderivative F. Then for any curve $\gamma : [a, b] \to U$ we have

$$\int_\gamma f(z)\ \mathrm{d}z = F(\gamma(b)) - F(\gamma(a)).$$

In particular, if γ is a *closed* curve (that is, $\gamma(a) = \gamma(b)$), then $\int_\gamma f(z)\ \mathrm{d}z = 0$.

Proof. Suppose first that γ is C^1. Then by the chain rule,

$$\int_\gamma f(z)\, \mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\, \mathrm{d}t = \int_a^b \frac{\mathrm{d}}{\mathrm{d}t}\left[F(\gamma(t))\right]\, \mathrm{d}t = F(\gamma(b)) - F(\gamma(a)),$$

where the last equality is the real fundamental theorem of calculus applied to the real and imaginary parts. If γ is only piecewise C^1, apply the above on each piece and note that the sum telescopes. $\square$

Note that the theorem *assumes* that an antiderivative F exists – it says nothing about when this happens. Whether or not every holomorphic function on U has an antiderivative turns out to depend delicately on the geometry of U, and this question is really the engine behind the next few sections. Let's do the fundamental example.

Example 2.14 (Integrating z^n around a Circle)
Fix $R > 0$ and an integer n, and let $\gamma(t) = Re^{2\pi i t}$ for $t \in [0, 1]$, the circle of radius R traversed once anticlockwise. We compute $\int_\gamma z^n\, \mathrm{d}z$.

If $n \neq -1$, then $z^{n+1}/(n+1)$ is an antiderivative of z^n on $\mathbb{C}\backslash\{0\}$ (or on $\mathbb{C}$ if $n \geq 0$), so since γ is closed,

$$\int_\gamma z^n\, \mathrm{d}z = 0.$$

If $n = -1$, we instead compute directly from the definition:

$$\int_\gamma \frac{\mathrm{d}z}{z} = \int_0^1 \frac{2\pi i Re^{2\pi i t}}{Re^{2\pi i t}}\, \mathrm{d}t = 2\pi i.$$

This example is small but its consequences are large. Since $2\pi i \neq 0$, the function $1/z$ has *no* antiderivative on any open set containing the circle $\{|z| = R\}$, for any $R > 0$. But if λ were a branch of the logarithm on $\mathbb{C}\backslash\{0\}$, then λ would be exactly such an antiderivative. So, as promised earlier, *there is no branch of the logarithm on* $\mathbb{C}\backslash\{0\}$.

Now, the fundamental theorem of calculus has a converse: if all integrals of f around closed curves vanish, then an antiderivative can be built.

Theorem 2.15 (Converse to the Fundamental Theorem of Calculus)
Let $U \subseteq \mathbb{C}$ be a domain and $f : U \to \mathbb{C}$ be continuous. If

$$\int_\gamma f(z)\, \mathrm{d}z = 0$$

for every closed curve γ in U, then f has an antiderivative on U.

Proof. Fix a base point $a_0 \in U$. Since U is path connected, for each $w \in U$ we can choose[a] a curve $\gamma_w : [0, 1] \to U$ from a_0 to w, and we define

$$F(w) = \int_{\gamma_w} f(z)\, \mathrm{d}z.$$

This is well defined, i.e. independent of the choice of curve: if γ_w and $\tilde{\gamma}_w$ are two such curves, then $\gamma_w + (-\tilde{\gamma}_w)$ is a closed curve in U, so by hypothesis

$$0 = \int_{\gamma_w + (-\tilde{\gamma}_w)} f(z)\,\mathrm{d}z = \int_{\gamma_w} f(z)\,\mathrm{d}z - \int_{\tilde{\gamma}_w} f(z)\,\mathrm{d}z.$$

We now check that F is differentiable with $F' = f$. Fix $w \in U$ and take $r > 0$ with $D(w, r) \subset U$. For $h \in \mathbb{C}$ with $0 < |h| < r$, let $\delta_h(t) = w + th$, $t \in [0, 1]$, be the radial segment from w to $w + h$. Then $\gamma_w + \delta_h + (-\gamma_{w+h})$ is a closed curve in U, so its integral vanishes, giving

$$F(w+h) = F(w) + \int_{\delta_h} f(z)\,\mathrm{d}z = F(w) + hf(w) + \int_{\delta_h} (f(z) - f(w))\,\mathrm{d}z,$$

where we used $\int_{\delta_h} \mathrm{d}z = h$. Hence by our basic estimate,

$$\left| \frac{F(w+h) - F(w)}{h} - f(w) \right| \le \frac{1}{|h|} \cdot \text{length}(\delta_h) \sup_{z \in \delta_h} |f(z) - f(w)| = \sup_{z \in \delta_h} |f(z) - f(w)|,$$

and this tends to 0 as $h \to 0$ by the continuity of f at w. So $F'(w) = f(w)$. □

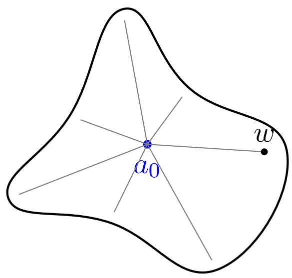
[a] Strictly, path connectedness gives a continuous curve, but for open sets one can always upgrade this to a piecewise C^1 curve, say by covering the curve with finitely many disks and taking line segments within them.

§2.3 Cauchy's Theorem for Star-Shaped Domains

We have just seen that 'all closed curve integrals vanish' is equivalent to having an antiderivative. The central theorem of this course, *Cauchy's theorem*, says that for a *holomorphic* function on a domain with reasonable geometry, all closed curve integrals really do vanish. In this section we prove this for *star-shaped* domains, which covers a surprising number of applications.

Definition 2.16 (Star-Shaped Domain)

A domain $U \subseteq \mathbb{C}$ is **star-shaped** if there is a point $a_0 \in U$ (a **centre**) such that for every $w \in U$, the straight line segment $[a_0, w]$ from a_0 to w lies entirely in U.



A star-shaped domain need not be convex (the domain above is not!), but every point can 'see' the centre along a straight line. Some quick observations, each of which is straightforward to check:

(i) every disk $D(a, r)$ is convex;

(ii) every convex domain is star-shaped (any point works as a centre);

(iii) every star-shaped domain is path connected.

None of the reverse implications hold in general. For instance, $\mathbb{C}\backslash\{x \in \mathbb{R} : x \leq 0\}$ is star-shaped (take $a_0 = 1$, say) but certainly not convex, and an annulus $\{1 < |z| < 2\}$ is path connected but not star-shaped.

The strategy for proving Cauchy's theorem on a star-shaped domain is to first establish it for the simplest possible closed curves – boundaries of triangles – and then to leverage the converse of the fundamental theorem of calculus.

Definition 2.17 (Triangle)
A **triangle** T in $\mathbb{C}$ is the convex hull of three points $z_1, z_2, z_3 \in \mathbb{C}$, that is

$$T = \{az_1 + bz_2 + cz_3 \mid a, b, c \geq 0,\ a + b + c = 1\}.$$

We write ∂T for the boundary of T, viewed as the closed curve which traverses the three sides of T once.

Corollary 2.18 (Antiderivatives on Star-Shaped Domains)
Let U be a star-shaped domain and $f : U \to \mathbb{C}$ be continuous with

$$\int_{\partial T} f(z)\ \mathrm{d}z = 0$$

for every triangle $T \subseteq U$. Then f has an antiderivative on U.

Proof. We repeat the proof of the converse to the fundamental theorem of calculus, being economical about which curves we use. Let a_0 be a centre of U, and for $w \in U$ define

$$F(w) = \int_{[a_0,w]} f(z)\ \mathrm{d}z,$$

integrating along the straight segment, which lies in U as U is star-shaped. Given $w \in U$, choose $r > 0$ with $D(w, r) \subset U$, and let $h \in \mathbb{C}$ with $0 < |h| < r$. The key point is that the closed curve $[a_0, w] + [w, w + h] + [w + h, a_0]$ is precisely the boundary of the triangle T with vertices $a_0, w, w + h$, and this triangle lies in U: its sides do, and every point of the segment $[w, w + h]$ lies in $D(w, r) \subseteq U$, so every segment from a_0 to a point of T lies in U by the star-shape property[a]. Hence by hypothesis

$$F(w + h) - F(w) = \int_{[w,w+h]} f(z)\ \mathrm{d}z,$$

and exactly as before, continuity of f gives $F'(w) = f(w)$. □

[a] To see $T \subset U$ carefully: every point of T lies on a segment from a_0 to a point of $[w, w + h]$, and $[w, w + h] \subset D(w, r) \subset U$.

So everything now rests on showing that integrals of holomorphic functions around triangles vanish. This is the technical heart of Cauchy's theorem, and the proof – due to Goursat – is a beautiful 'divide and conquer' argument.

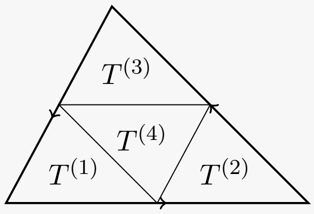
Theorem 2.19 (Cauchy's Theorem for Triangles)
Let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be holomorphic. Then

$$\int_{\partial T} f(z)\, \mathrm{d}z = 0$$

for every triangle $T \subseteq U$.

Proof. For a triangle T, write $\eta(T) = \int_{\partial T} f(z)\, \mathrm{d}z$; our goal is $\eta(T) = 0$.

Subdivide T into four smaller triangles $T^{(1)}, \dots, T^{(4)}$ by joining the midpoints of its sides, as in the picture below.



Traversing each $\partial T^{(j)}$ anticlockwise, the integrals over the three interior sides cancel in pairs (each is traversed once in each direction), so

$$\eta(T) = \sum_{j=1}^{4} \eta(T^{(j)}).$$

By the triangle inequality, there is some j with $|\eta(T^{(j)})| \geq |\eta(T)|/4$. Set $T_0 = T$ and $T_1 = T^{(j)}$, and repeat this construction inductively to obtain nested triangles $T_0 \supset T_1 \supset T_2 \supset \cdots$ with

$$|\eta(T_n)| \geq \frac{|\eta(T_0)|}{4^n}, \quad \text{and} \quad \operatorname{length}(\partial T_n) = \frac{\operatorname{length}(\partial T_0)}{2^n},$$

the latter since each subdivision halves all side lengths.

The T_n are nested non-empty compact sets with diameters tending to zero, so their intersection is a single point: $\bigcap_{n \geq 0} T_n = \{z_0\}$, with $z_0 \in T \subseteq U$.

Now we finally use that f is differentiable at z_0. Given $\varepsilon > 0$, there is $\delta > 0$ such that $D(z_0, \delta) \subseteq U$ and

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0| \quad \text{for all } z \in D(z_0, \delta).$$

Also, since the affine function $z \mapsto f(z_0) + f'(z_0)(z - z_0)$ has an antiderivative, its integral around any closed curve vanishes, so for every n,

$$\eta(T_n) = \int_{\partial T_n} \big(f(z) - f(z_0) - f'(z_0)(z - z_0)\big)\, \mathrm{d}z.$$

Choose n large enough that $T_n \subseteq D(z_0, \delta)$. Then for $z \in \partial T_n$ we have $|z - z_0| \leq \operatorname{length}(\partial T_n)$, so estimating the integral,

$$\frac{|\eta(T_0)|}{4^n} \leq |\eta(T_n)| \leq \varepsilon \operatorname{length}(\partial T_n)^2 = \frac{\varepsilon \operatorname{length}(\partial T_0)^2}{4^n}.$$

Hence $|\eta(T_0)| \leq \varepsilon \operatorname{length}(\partial T_0)^2$ for every $\varepsilon > 0$, forcing $\eta(T_0) = 0$. □

For later applications (in particular when we prove the Cauchy integral formula) it is extremely useful to relax the hypothesis slightly, allowing f to fail to be differentiable at finitely many points, so long as it remains continuous there.

Theorem 2.20 (Cauchy's Theorem for Triangles, with Exceptional Points)
Let $U \subseteq \mathbb{C}$ be open, $S \subset U$ be finite, and let $f : U \to \mathbb{C}$ be continuous on U and holomorphic on $U \backslash S$. Then $\int_{\partial T} f(z)\, \mathrm{d}z = 0$ for every triangle $T \subseteq U$.

Proof. Subdivide T as before, but n times over, keeping *all* of the 4^n small triangles $T_1, \ldots, T_N$ $(N = 4^n)$. Again the interior sides cancel, so

$$\int_{\partial T} f(z)\, \mathrm{d}z = \sum_{j=1}^{N} \int_{\partial T_j} f(z)\, \mathrm{d}z.$$

By the previous theorem, $\int_{\partial T_j} f(z)\, \mathrm{d}z = 0$ for every T_j that avoids S. Now, each point of S can lie in at most 6 of the small triangles (a point can be a vertex of at most 6 of them), and each small triangle has $\operatorname{length}(\partial T_j) = 2^{-n} \operatorname{length}(\partial T)$. Also f, being continuous on the compact set T, is bounded there, say $|f| \leq M$ on T. Hence

$$\left| \int_{\partial T} f(z)\, \mathrm{d}z \right| \leq 6|S|M \cdot \frac{\operatorname{length}(\partial T)}{2^n},$$

and letting $n \to \infty$ gives the result. □

Combining our work, we obtain the form of Cauchy's theorem that we will use constantly.

Theorem 2.21 (Cauchy's Theorem for Star-Shaped Domains)
Let U be a star-shaped domain, and let $f : U \to \mathbb{C}$ be continuous on U and holomorphic on $U \backslash S$ for some finite set S. Then

$$\int_{\gamma} f(z)\, \mathrm{d}z = 0$$

for every closed curve γ in U. In particular this holds when U is convex, and when f is holomorphic on all of U.

Proof. By the previous theorem, $\int_{\partial T} f(z)\, \mathrm{d}z = 0$ for every triangle $T \subseteq U$. Since U is star-shaped, our corollary gives that f has an antiderivative on U, and then the fundamental theorem of calculus gives $\int_{\gamma} f(z)\, \mathrm{d}z = 0$ for every closed γ. □

Remark (Deforming Curves)**.** The reader should think of Cauchy's theorem as a statement about the *geometry* of U as much as about f. The correct general statement, which we will prove later in the course, is that $\int_\gamma f \, \mathrm{d}z = 0$ whenever the closed curve γ can be continuously shrunk to a point within U (that is, γ is *null-homotopic*), and more generally that $\int_\gamma f \, \mathrm{d}z$ is unchanged when γ is continuously deformed within U. Star-shaped domains are exactly domains where every closed curve can be shrunk to the centre, sliding along straight lines. What can go wrong in general is visible already for $f(z) = 1/z$ on $\mathbb{C}\setminus\{0\}$: a circle around the origin cannot be shrunk to a point without crossing 0, and correspondingly its integral is $2\pi i \neq 0$.

§3 Cauchy's Integral Formula and its Applications

§3.1 The Cauchy Integral Formula

We now come to a truly remarkable result: the value of a holomorphic function at a point is completely determined by its values on a circle around that point, via an explicit integral formula. Almost all of the 'magic' properties of holomorphic functions flow from this.

Throughout, for a disk $D(a, \rho)$ we write $\int_{\partial D(a,\rho)} f(z) \, \mathrm{d}z$ for $\int_\gamma f(z) \, \mathrm{d}z$ where $\gamma(t) = a + \rho e^{2\pi i t}$, $t \in [0, 1]$, the boundary circle traversed once anticlockwise.

We will need a simple lemma about swapping limits and integrals.

Lemma 3.1 (Uniform Convergence and Integrals)
Let $\gamma : [a, b] \to \mathbb{C}$ be a curve and let f_n be a sequence of continuous functions on the image of γ, converging uniformly on the image of γ to f. Then

$$\int_\gamma f_n(z) \, \mathrm{d}z \to \int_\gamma f(z) \, \mathrm{d}z.$$

Proof. By our basic estimate,

$$\left| \int_\gamma f_n(z) \, \mathrm{d}z - \int_\gamma f(z) \, \mathrm{d}z \right| \le \operatorname{length}(\gamma) \sup_\gamma |f_n - f| \to 0.$$

□

Theorem 3.2 (Cauchy Integral Formula for a Disk)
Let $D = D(a, r)$ and let $f : D \to \mathbb{C}$ be holomorphic. Then for any $0 < \rho < r$ and any $w \in D(a, \rho)$,

$$f(w) = \frac{1}{2\pi i} \int_{\partial D(a,\rho)} \frac{f(z)}{z - w} \, \mathrm{d}z.$$

Proof. Fix $w \in D(a, \rho)$ and define $h : D \to \mathbb{C}$ by

$$h(z) = \begin{cases} \frac{f(z) - f(w)}{z - w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w. \end{cases}$$

Then h is continuous on D and holomorphic on $D\backslash\{w\}$. Since D is convex, Cauchy's theorem (with one exceptional point) gives

$$0 = \int_{\partial D(a,\rho)} h(z)\,\mathrm{d}z = \int_{\partial D(a,\rho)} \frac{f(z)}{z-w}\,\mathrm{d}z - f(w)\int_{\partial D(a,\rho)} \frac{\mathrm{d}z}{z-w}.$$

So it suffices to prove that

$$\int_{\partial D(a,\rho)} \frac{\mathrm{d}z}{z-w} = 2\pi i.$$

For $z \in \partial D(a,\rho)$ we have $|w-a| < \rho = |z-a|$, so expanding as a geometric series,

$$\frac{1}{z-w} = \frac{1}{(z-a)\left(1-\frac{w-a}{z-a}\right)} = \sum_{j=0}^{\infty} \frac{(w-a)^j}{(z-a)^{j+1}},$$

with the convergence uniform in $z \in \partial D(a,\rho)$ by the Weierstrass M-test (the terms are bounded by $(|w-a|/\rho)^j/\rho$). Hence by our lemma we may integrate term by term:

$$\int_{\partial D(a,\rho)} \frac{\mathrm{d}z}{z-w} = \sum_{j=0}^{\infty}(w-a)^j \int_{\partial D(a,\rho)} \frac{\mathrm{d}z}{(z-a)^{j+1}}.$$

For $j \geq 1$ the integrand $(z-a)^{-j-1}$ has an antiderivative near the circle, so those terms vanish; and for $j=0$ we computed the integral directly before, obtaining $2\pi i$. This completes the proof. □

Taking $w = a$ and parameterising the circle explicitly, we obtain a particularly pleasant special case.

Corollary 3.3 (Mean Value Property)

If $f : D(a,r) \to \mathbb{C}$ is holomorphic and $0 < \rho < r$, then

$$f(a) = \int_0^1 f(a + \rho e^{2\pi i t})\,\mathrm{d}t.$$

In other words – the value of a holomorphic function at a point is the *average* of its values on any circle centred at that point.

§3.2 Liouville's Theorem and the Fundamental Theorem of Algebra

Here is the first of the surprising rigidity theorems that we promised back in the first section.

Theorem 3.4 (Liouville's Theorem)

Let $f : \mathbb{C} \to \mathbb{C}$ be entire and bounded. Then f is constant.

Proof. Say $|f| \leq M$ on $\mathbb{C}$. Fix $w \in \mathbb{C}$, and let $\rho > 2|w|$. By the Cauchy integral formula applied on a large disk $D(0,\rho')$ with $\rho < \rho'$,

$$f(w) - f(0) = \frac{1}{2\pi i}\int_{\partial D(0,\rho)} f(z)\left(\frac{1}{z-w} - \frac{1}{z}\right) \mathrm{d}z = \frac{1}{2\pi i}\int_{\partial D(0,\rho)} \frac{wf(z)}{z(z-w)} \mathrm{d}z.$$

For z on the circle $|z| = \rho$ we have $|z-w| \geq \rho - |w| \geq \rho/2$, so estimating,

$$|f(w) - f(0)| \leq \frac{1}{2\pi} \cdot 2\pi\rho \cdot \frac{|w|M}{\rho \cdot \rho/2} = \frac{2|w|M}{\rho}.$$

Letting $\rho \to \infty$ gives $f(w) = f(0)$. Since w was arbitrary, f is constant. □

Notice how strong this is: there is no entire analogue of sin on $\mathbb{R}$, bounded and wiggling forever – indeed sin itself is wildly unbounded on $\mathbb{C}$ (consider $\sin(iy)$). One spectacular application is a genuinely short proof of the fundamental theorem of algebra.

Theorem 3.5 (Fundamental Theorem of Algebra)
Every non-constant polynomial with complex coefficients has a root in $\mathbb{C}$.

Proof. Let $p(z) = a_n z^n + \cdots + a_1 z + a_0$ with $n \geq 1$ and $a_n \neq 0$. For $z \neq 0$,

$$|p(z)| \geq |z|^n \left(|a_n| - \frac{|a_{n-1}|}{|z|} - \cdots - \frac{|a_0|}{|z|^n}\right),$$

and the bracket tends to $|a_n| > 0$ as $|z| \to \infty$. So there is $R > 0$ such that $|p(z)| \geq 1$ for all $|z| > R$.

Now suppose for contradiction that p has no root. Then $g = 1/p$ is entire. By the above, $|g| \leq 1$ outside $D(0,R)$; and $|g|$ is continuous on the compact set $\overline{D(0,R)}$, hence bounded there too. So g is a bounded entire function, and by Liouville's theorem g is constant – but then p is constant, a contradiction. □

Remark. It follows by induction (dividing out roots) that every polynomial of degree $n \geq 1$ factors completely as $p(z) = a_n(z - z_1)\cdots(z - z_n)$. So $\mathbb{C}$ is algebraically closed – the analytic input of Liouville's theorem proves a purely algebraic statement.

§3.3 Taylor Expansions

We now show that holomorphic functions are *analytic*: locally, they are given by convergent power series. Together with our earlier work on power series, this instantly upgrades 'differentiable once' to 'differentiable infinitely often'.

Theorem 3.6 (Taylor Expansion)
Let $f : D(a,R) \to \mathbb{C}$ be holomorphic. Then f has a convergent power series representation on $D(a,R)$:

$$f(w) = \sum_{n=0}^{\infty} c_n (w-a)^n, \quad \text{where} \quad c_n = \frac{1}{2\pi i}\int_{\partial D(a,\rho)} \frac{f(z)}{(z-a)^{n+1}} \mathrm{d}z$$

for any $0 < \rho < R$.

Proof. Let $0 < \rho < R$ and $w \in D(a, \rho)$. By the Cauchy integral formula and the geometric series expansion from its proof (which converges uniformly for $z \in \partial D(a, \rho)$),

$$\begin{aligned} f(w) &= \frac{1}{2\pi i} \int_{\partial D(a,\rho)} \frac{f(z)}{z-w}\, \mathrm{d}z = \frac{1}{2\pi i} \int_{\partial D(a,\rho)} f(z) \sum_{n=0}^{\infty} \frac{(w-a)^n}{(z-a)^{n+1}}\, \mathrm{d}z \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial D(a,\rho)} \frac{f(z)}{(z-a)^{n+1}}\, \mathrm{d}z \right) (w-a)^n, \end{aligned}$$

where the interchange of sum and integral is justified by our uniform convergence lemma (f is bounded on the circle, so the partial sums converge uniformly there).

Write $c_n(\rho)$ for the coefficient above. We have shown $f(w) = \sum c_n(\rho)(w-a)^n$ on $D(a, \rho)$, so by our theorem on derivatives of power series, $c_n(\rho) = f^{(n)}(a)/n!$. In particular $c_n(\rho)$ does not depend on ρ, so we may write c_n unambiguously. □

Corollary 3.7 (Holomorphic Functions are Infinitely Differentiable)

If f is holomorphic on an open set $U \subseteq \mathbb{C}$, then f has derivatives of all orders on U, and they are all holomorphic on U.

Proof. Around any point of U there is a disk on which f is a convergent power series, and power series have derivatives of all orders inside their disk of convergence. □

This settles the promise made at the start of the course, and it also completes the picture for the Cauchy-Riemann equations: if $f = u + iv$ is holomorphic then u, v are automatically C^∞, and in particular the harmonic function remark from Section 1 applies in earnest.

Remark (Holomorphic = Analytic)**.** We say f is **analytic** at a if it is given by a convergent power series in some neighbourhood of a. Combining the Taylor expansion theorem with our theorem on differentiating power series, the following are equivalent for a function on an open set U: f is holomorphic on U; f is analytic at every point of U; f has complex derivatives of all orders on U. Nothing like this is true on $\mathbb{R}$: the function $x \mapsto e^{-1/x^2}$ (extended by 0) is C^∞ but is not given by its Taylor series at 0.

From the Taylor coefficients we can also read off integral formulas for all derivatives, along with very useful bounds.

Theorem 3.8 (Cauchy Integral Formula for Derivatives & Cauchy Estimates)

Let $f : D(a, R) \to \mathbb{C}$ be holomorphic. Then for any $0 < \rho < R$ and $w \in D(a, \rho)$,

$$f^{(k)}(w) = \frac{k!}{2\pi i} \int_{\partial D(a,\rho)} \frac{f(z)}{(z-w)^{k+1}}\, \mathrm{d}z.$$

Moreover, we have the estimate

$$\sup_{w \in D(a,R/2)} |f^{(k)}(w)| \le \frac{k!\, 2^{k+1}}{R^k} \sup_{z \in D(a,R)} |f(z)|.$$

Proof. The case $k = 0$ is the Cauchy integral formula. For $k = 1$, consider $g(z) = f(z)/(z-w)$, holomorphic on $D(a,R)\backslash\{w\}$, with

$$g'(z) = \frac{f'(z)}{z-w} - \frac{f(z)}{(z-w)^2}.$$

Now g is an antiderivative of g' on $D(a,R)\backslash\{w\}$, which contains the circle $\partial D(a,\rho)$, so $\int_{\partial D(a,\rho)} g'(z)\, \mathrm{d}z = 0$ by the fundamental theorem of calculus. Hence

$$\int_{\partial D(a,\rho)} \frac{f'(z)}{z-w}\, \mathrm{d}z = \int_{\partial D(a,\rho)} \frac{f(z)}{(z-w)^2}\, \mathrm{d}z,$$

and applying the Cauchy integral formula to the holomorphic function f' gives the $k = 1$ case. For general k we induct: assuming the formula for k (for all holomorphic functions on $D(a,R)$), the same trick with $g(z) = f(z)/(z-w)^{k+1}$ gives

$$\int_{\partial D(a,\rho)} \frac{f'(z)}{(z-w)^{k+1}}\, \mathrm{d}z = (k+1)\int_{\partial D(a,\rho)} \frac{f(z)}{(z-w)^{k+2}}\, \mathrm{d}z,$$

and applying the inductive hypothesis to f' we get

$$f^{(k+1)}(w) = \frac{k!}{2\pi i}\int_{\partial D(a,\rho)} \frac{f'(z)}{(z-w)^{k+1}}\, \mathrm{d}z = \frac{(k+1)!}{2\pi i}\int_{\partial D(a,\rho)} \frac{f(z)}{(z-w)^{k+2}}\, \mathrm{d}z.$$

For the estimate, we may assume $M = \sup_{D(a,R)} |f| < \infty$. Let $\rho \in (R/2, R)$ and $w \in D(a, R/2)$. Then $|z-w| \geq \rho - R/2$ for $z \in \partial D(a,\rho)$, so

$$|f^{(k)}(w)| \leq \frac{k!}{2\pi} \cdot 2\pi\rho \cdot \frac{M}{(\rho - R/2)^{k+1}} = \frac{k!\rho M}{(\rho-R/2)^{k+1}},$$

and letting $\rho \to R$ gives $|f^{(k)}(w)| \leq k!RM/(R/2)^{k+1} = k!2^{k+1}M/R^k$. □

We can also now prove a converse to Cauchy's theorem, which gives a criterion for holomorphy requiring no differentiability assumptions whatsoever.

Theorem 3.9 (Morera's Theorem)
Let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be continuous such that

$$\int_\gamma f(z)\, \mathrm{d}z = 0$$

for every closed curve γ in U. Then f is holomorphic on U.

Proof. The statement is local, so we may assume U is a disk (restricting to a disk around each point, where the hypothesis still holds). Then by the converse to the fundamental theorem of calculus, f has an antiderivative F on U, i.e. $F' = f$. But F is holomorphic, so it is infinitely differentiable, and in particular $f = F'$ is holomorphic. □

Corollary 3.10 (Removing Exceptional Points)

Let $U \subseteq \mathbb{C}$ be open, $S \subset U$ finite, and let $f : U \to \mathbb{C}$ be continuous on U and holomorphic on $U \backslash S$. Then f is holomorphic on U.

Proof. Around each point $a \in U$ take a disk $D(a, r) \subseteq U$. By Cauchy's theorem for star-shaped domains (with exceptional points), every closed curve integral of f in $D(a, r)$ vanishes, so by Morera's theorem f is holomorphic on $D(a, r)$. □

§3.4 Zeros and the Identity Theorem

Power series give us very tight control over the zeros of holomorphic functions. Suppose $f : D(a, R) \to \mathbb{C}$ is holomorphic and not identically zero, and write its Taylor expansion $f(z) = \sum_{n \geq 0} c_n(z-a)^n$. Since $f \not\equiv 0$, some coefficient is non-zero (by our power series theorem, if all $c_n = 0$ then $f \equiv 0$ on the disk), so we may let

$$m = \min\{n \geq 0 \mid c_n \neq 0\}.$$

Then we can factor

$$f(z) = (z-a)^m g(z), \quad \text{where} \quad g(z) = \sum_{n=m}^{\infty} c_n(z-a)^{n-m}$$

is holomorphic on $D(a, R)$ with $g(a) = c_m \neq 0$.

Definition 3.11 (Order of a Zero)

In the above setting, if $f(a) = 0$ (that is, $m \geq 1$), we say f has a **zero of order m** at a. Equivalently, m is the least n with $f^{(n)}(a) \neq 0$. A zero of order 1 is called a **simple zero**.

Theorem 3.12 (Principle of Isolated Zeros)

Let $f : D(a, R) \to \mathbb{C}$ be holomorphic and not identically zero. Then there is some $0 < r \leq R$ such that $f(z) \neq 0$ for all z with $0 < |z-a| < r$.

Proof. If $f(a) \neq 0$, this is immediate from continuity. If $f(a) = 0$, write $f(z) = (z-a)^m g(z)$ as above with $g(a) \neq 0$. By continuity of g there is $r > 0$ with $g \neq 0$ on $D(a, r)$, and then $f(z) \neq 0$ for $0 < |z-a| < r$. □

In other words – the zeros of a non-trivial holomorphic function are *isolated*: each zero has a punctured neighbourhood free of other zeros. There is no holomorphic function vanishing on a little segment, or on a little disk, other than the zero function. This has a striking global consequence.

Theorem 3.13 (Identity Theorem)

Let $U \subseteq \mathbb{C}$ be a domain and $f, g : U \to \mathbb{C}$ be holomorphic. If the set $S = \{z \in U \mid$

$f(z) = g(z)\}$ contains a non-isolated point[a], then $f = g$ on all of U.

[a]A point $w \in S$ is *non-isolated* if every punctured disk $D(w, r)\backslash\{w\}$ meets S.

Proof. Let $h = f - g$, so $S = \{z \in U \mid h(z) = 0\}$, and suppose $w \in S$ is a non-isolated point of S. Define

$$U_0 = \{z \in U \mid h \equiv 0 \text{ on some } D(z, r)\},$$
$$U_1 = \{z \in U \mid h^{(n)}(z) \neq 0 \text{ for some } n \geq 0\}.$$

These are disjoint: if h vanishes identically near z then so do all its derivatives at z. They also cover U: if $z \notin U_1$ then every Taylor coefficient of h at z vanishes, and so $h \equiv 0$ on any disk $D(z, r) \subseteq U$, meaning $z \in U_0$.

Moreover both sets are open: U_0 is open essentially by definition, and U_1 is open since each $h^{(n)}$ is continuous. Since U is a domain (hence connected), one of U_0, U_1 is empty.

Finally, the non-isolated point w lies in U_0: if it did not, then $h \not\equiv 0$ near w, so by the principle of isolated zeros h would be non-vanishing on some punctured disk around w, contradicting non-isolation. So $U_0 \neq \emptyset$, hence $U_1 = \emptyset$ and $U = U_0$, that is, $h \equiv 0$ on U. □

Remark (Analytic Continuation)**.** The identity theorem says that a holomorphic function on a domain is completely determined by its values on any set with a limit point – say, a tiny disk, or a convergent sequence. If $U \subseteq V$ are domains and $f : U \to \mathbb{C}$ is holomorphic, then there is *at most one* holomorphic $g : V \to \mathbb{C}$ with $g|_U = f$; such a g is called the **analytic continuation** of f to V. It may or may not exist! For instance $f(z) = \sum_{n \geq 0} z^n$ is holomorphic on $D(0, 1)$, and admits the analytic continuation $1/(1 - z)$ to $\mathbb{C}\backslash\{1\}$, far beyond the disk of convergence. In contrast, one can show that $f(z) = \sum_{n \geq 0} z^{n!}$ (also holomorphic on $D(0, 1)$) admits no analytic continuation past the unit disk whatsoever.

Example 3.14 (Real Identities Propagate)

Since $\sin^2 x + \cos^2 x = 1$ holds for all real x, the holomorphic function $\sin^2 z + \cos^2 z - 1$ vanishes on $\mathbb{R}$, a set with (many!) non-isolated points. By the identity theorem it vanishes identically, so $\sin^2 z + \cos^2 z = 1$ for *all* $z \in \mathbb{C}$. The same argument transfers essentially any polynomial identity from $\mathbb{R}$ to $\mathbb{C}$.

§3.5 The Maximum Modulus Principle

Another consequence of the mean value property is that a holomorphic function can never have an interior maximum of modulus, unless it is constant.

Theorem 3.15 (Local Maximum Modulus Principle)

Let $f : D(a, R) \to \mathbb{C}$ be holomorphic. If $|f(z)| \leq |f(a)|$ for all $z \in D(a, R)$, then f is constant.

Proof. Let $0 < \rho < R$. By the mean value property and the basic estimate,

$$|f(a)| = \left| \int_0^1 f(a + \rho e^{2\pi it}) \, \mathrm{d}t \right| \le \sup_{t \in [0,1]} |f(a + \rho e^{2\pi it})| \le |f(a)|,$$

using the hypothesis for the final inequality. So both inequalities are equalities. Equality in the basic estimate forces $t \mapsto f(a+\rho e^{2\pi it})$ to be constant, with modulus $|f(a)|$ (by the first equality). Since $\rho \in (0, R)$ was arbitrary, $|f| \equiv |f(a)|$ is constant on $D(a, R)$.

Finally, a holomorphic function of constant modulus is constant. If $|f|^2 = u^2 + v^2$ is a non-zero constant, differentiating gives $uu_x + vv_x = 0$ and $uu_y + vv_y = 0$; using the Cauchy-Riemann equations to eliminate the y-derivatives, $uu_x - vu_y = 0$ and $uu_y + vu_x = 0$, and solving this linear system (whose determinant is $u^2+v^2 \neq 0$) gives $u_x = u_y = 0$, and similarly for v. So $f' = 0$ and f is constant on the (connected) disk. If instead $|f| \equiv 0$ then $f \equiv 0$. □

Corollary 3.16 (Maximum Modulus Principle)

Let U be a bounded open set and let $f : \overline{U} \to \mathbb{C}$ be continuous on $\overline{U}$ and holomorphic on U. Then $|f|$ attains its maximum over $\overline{U}$ on the boundary ∂U.

Proof. Since $\overline{U}$ is compact and $|f|$ continuous, the maximum is attained at some $w \in \overline{U}$. If $w \in \partial U$ we are done, so suppose $w \in U$, and take $D(w, r) \subseteq U$. By the local maximum modulus principle, f is constant on $D(w, r)$, and hence by the identity theorem f is constant on the connected component U' of U containing w. By continuity f is constant on $\overline{U'}$, so the value $|f(w)|$ is also attained at points of $\partial U' \subseteq \partial U$.[a] □

[a] To see $\partial U' \subseteq \partial U$: a boundary point z of U' cannot lie in U, for otherwise a small disk around z in U would be connected to points of U', forcing $z \in U'$, which is impossible as U' is open. So $z \in \overline{U} \backslash U = \partial U$.

§4 Uniform Limits of Holomorphic Functions

We now look at sequences of holomorphic functions that converge uniformly – or rather, uniformly on the parts of the domain that matter.

Definition 4.1 (Locally Uniform Convergence)

Let $U \subset \mathbb{C}$ be open, and let $f_n : U \to \mathbb{C}$ be a sequence of functions. We say that f_n converges **locally uniformly** on U if for each $a \in U$ there is some $r > 0$ such that f_n converges uniformly on $D(a, r)$.

Example 4.2 (Sequence of Functions Converging Locally Uniformly)

Consider the sequence of functions $f_n(z) = z^n$. Then $f_n \to 0$ locally uniformly on $D(0, 1)$, but not uniformly on $D(0, 1)$. This is because $f_n \to 0$ uniformly on any disk $D(0, r)$ if $r < 1$, but $\sup_{z \in D(0,1)} |f_n(z)| = 1$ for each n.

Proposition 4.3

The sequence of functions f_n converges locally uniformly on U if and only if f_n converges uniformly on each compact subset $K \subset U$.

Proof. For the forward direction, suppose f_n converges locally uniformly, and let $K \subset U$ be compact. Each $a \in K$ has a disk $D(a, r_a)$ on which f_n converges uniformly, and these disks cover K, so by compactness finitely many of them, say $D(a_1, r_{a_1}), \dots, D(a_m, r_{a_m})$, cover K. Given $\varepsilon > 0$, for each i there is N_i such that $|f_n(z) - f(z)| < \varepsilon$ for all $n \geq N_i$ and $z \in D(a_i, r_{a_i})$; taking $N = \max_i N_i$ handles all $z \in K$ at once. So f_n converges uniformly on K.

Conversely, for each $a \in U$ there is $r > 0$ with the compact disk $\overline{D(a,r)} \subset U$, and uniform convergence on $\overline{D(a,r)}$ implies uniform convergence on $D(a,r)$. $\square$

Theorem 4.4 (Uniform Limits of Holomorphic Functions)

Let $U \subset \mathbb{C}$ be open, and $f_n : U \to \mathbb{C}$ be a sequence of holomorphic functions. If f_n converges locally uniformly on U to some function $f : U \to \mathbb{C}$, then f is holomorphic. Moreover, $f_n' \to f'$ locally uniformly on U[a].

[a] By applying this result iteratively, we also get $f_n^{(k)} \to f^{(k)}$ locally uniformly on U, for every k.

Proof. Let $a \in U$, and choose $r > 0$ such that $\overline{D(a,r)} \subset U$ and $f_n \to f$ uniformly on $\overline{D(a,r)}$. A uniform limit of continuous functions is continuous, so f is continuous on $\overline{D(a,r)}$.

Now let γ be any closed curve in $D(a,r)$. The disk $D(a,r)$ is convex, so Cauchy's theorem gives $\int_\gamma f_n(z)\, \mathrm{d}z = 0$ for every n. By uniform convergence on the image of γ (a subset of $D(a,r)$), our lemma on swapping limits and integrals gives

$$\int_\gamma f(z)\, \mathrm{d}z = \lim_{n\to\infty} \int_\gamma f_n(z)\, \mathrm{d}z = 0.$$

Then Morera's theorem shows that f is holomorphic on $D(a,r)$, and since $a \in U$ was arbitrary, f is holomorphic on U.

For the convergence of derivatives, again take $a \in U$ and r as above. Applying the Cauchy estimate (with $k = 1$, $R = r$) to the holomorphic function $f_n - f$,

$$\sup_{z \in D(a, r/2)} |f_n'(z) - f'(z)| \leq \frac{4}{r} \sup_{z \in D(a,r)} |f_n(z) - f(z)| \to 0$$

as $n \to \infty$. So $f_n' \to f'$ uniformly on $D(a, r/2)$, giving locally uniform convergence. $\square$

This result spectacularly fails for real functions, as seen by the following theorem (which we will not prove).

Theorem 4.5 (Weierstrass Approximation Theorem)

Let $f : [a,b] \to \mathbb{R}$ be a continuous function on a compact interval $[a,b] \subset \mathbb{R}$. Then there is a sequence of polynomials converging to f uniformly on $[a, b]$.

Since there are continuous nowhere-differentiable functions on $[a, b]$, a uniform limit of real polynomials – the nicest functions imaginable – need not be differentiable at even a single point. For holomorphic functions, by contrast, local uniform limits preserve *everything*.

§5 Winding Numbers and the General Cauchy Theorem

So far, Cauchy's theorem needs the domain to be star-shaped. To go further we need a way of measuring how a closed curve sits inside a general domain – in particular, how many times it wraps around any given point. This is the *winding number*, and with it we can state and prove Cauchy's theorem in essentially complete generality.

§5.1 Winding Numbers

Let $\gamma : [a, b] \to \mathbb{C}$ be a closed curve and let w be a point not on the curve. For each t we can certainly write

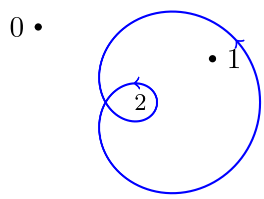
$$\gamma(t) = w + r(t)e^{i\theta(t)},$$

where $r(t) = |\gamma(t) - w| > 0$ is uniquely determined and continuous. The angle $\theta(t)$ is only determined up to adding multiples of 2π – but as we will show below, it is possible to make a *continuous* choice of θ, and this is what makes the following definition sensible.

Definition 5.1 (Winding Number)
Let $\gamma : [a, b] \to \mathbb{C}$ be a closed curve, $w \notin \text{image}(\gamma)$, and let $\theta : [a, b] \to \mathbb{R}$ be a continuous function with $\gamma(t) = w + r(t)e^{i\theta(t)}$. The **winding number** (or **index**) of γ about w is

$$I(\gamma; w) = \frac{\theta(b) - \theta(a)}{2\pi}.$$

Two things need checking. First, since $\gamma(a) = \gamma(b)$, we have $e^{i(\theta(b)-\theta(a))} = 1$, so $I(\gamma; w)$ is an *integer*. Second, it is well defined: if θ_1 is another continuous choice, then $(\theta_1 - \theta)/2\pi$ is a continuous integer-valued function on an interval, hence constant, so both choices give the same value of $I(\gamma; w)$.



The picture shows a closed curve along with the winding number of the curve about points of the various regions of the complement – points it wraps around once, twice, and not at all.

For the curves we care about (piecewise C^1), a continuous angle function always exists, and better still, the winding number is computed by an integral.

Lemma 5.2 (Existence of Continuous Argument)
Let $\gamma : [a,b] \to \mathbb{C}\backslash\{w\}$ be a (piecewise C^1) curve. Then there is a piecewise C^1 function $\theta : [a,b] \to \mathbb{R}$ with $\gamma(t) = w + r(t)e^{i\theta(t)}$, where $r(t) = |\gamma(t) - w|$. Moreover, if γ is closed, then

$$I(\gamma; w) = \frac{1}{2\pi i} \int_\gamma \frac{\mathrm{d}z}{z - w}.$$

Proof. Define

$$h(t) = \int_a^t \frac{\gamma'(s)}{\gamma(s) - w}\,\mathrm{d}s.$$

The integrand is bounded and continuous away from the finitely many points where γ' jumps, so h is continuous and piecewise C^1, with $h'(t) = \gamma'(t)/(\gamma(t) - w)$ away from those points. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[(\gamma(t) - w)e^{-h(t)}\right] = \gamma'(t)e^{-h(t)} - (\gamma(t) - w)h'(t)e^{-h(t)} = 0$$

except at finitely many points, so by continuity $(\gamma(t) - w)e^{-h(t)}$ is constant, equal to $\gamma(a) - w$. Writing $\gamma(a) - w = r(a)e^{i\alpha}$, we get

$$\gamma(t) - w = r(a)e^{\operatorname{Re} h(t)}e^{i(\alpha + \operatorname{Im} h(t))},$$

so $\theta(t) = \alpha + \operatorname{Im} h(t)$ is a continuous (piecewise C^1) argument function, and comparing moduli, $r(t) = r(a)e^{\operatorname{Re} h(t)}$.

Now suppose γ is closed. Then $r(b) = r(a)$, so $\operatorname{Re} h(b) = 0$, and

$$I(\gamma; w) = \frac{\theta(b) - \theta(a)}{2\pi} = \frac{\operatorname{Im} h(b)}{2\pi} = \frac{h(b)}{2\pi i} = \frac{1}{2\pi i}\int_a^b \frac{\gamma'(s)}{\gamma(s) - w}\,\mathrm{d}s = \frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}z}{z - w}.$$

□

Proposition 5.3 (Winding Numbers are Locally Constant)
Let $\gamma : [a,b] \to \mathbb{C}$ be a closed curve. Then $w \mapsto I(\gamma; w)$ is locally constant on $\mathbb{C}\backslash \operatorname{image}(\gamma)$; in particular, it is constant on each connected component of this set.

Proof. The function $w \mapsto \frac{1}{2\pi i}\int_\gamma \frac{\mathrm{d}z}{z-w}$ is continuous on the open set $\mathbb{C}\backslash \operatorname{image}(\gamma)$ (the integrand depends continuously on w, uniformly for z on the curve, since $|z - w|$ is bounded below near any fixed w_0). A continuous integer-valued function is locally constant, and a locally constant function is constant on connected components. □

Proposition 5.4 (Winding Number Far Away)
Let $\gamma : [a,b] \to \mathbb{C}$ be a closed curve. Then $\mathbb{C}\backslash \operatorname{image}(\gamma)$ has exactly one unbounded connected component Ω, and $I(\gamma; w) = 0$ for all $w \in \Omega$.

Proof. The image of γ is compact, so it lies in some disk $D(0, R)$. The connected set $\mathbb{C} \backslash D(0, R)$ lies in a single component Ω of the complement of the curve, which is then unbounded; every other component avoids $\mathbb{C} \backslash D(0, R)$ and hence is contained in $D(0, R)$, i.e. is bounded. Finally, if $w \in \mathbb{C} \backslash D(0, R)$, then $z \mapsto 1/(z - w)$ is holomorphic on the convex set $D(0, R)$ which contains γ, so $I(\gamma; w) = 0$ by Cauchy's theorem for convex domains. By local constancy, $I(\gamma; \cdot) = 0$ on all of Ω. □

§5.2 The General Cauchy Theorem

We can now state the theorem. The correct condition on the curve turns out to be the following.

Definition 5.5 (Homologous to Zero)
Let $U \subseteq \mathbb{C}$ be open. A closed curve γ in U is **homologous to zero** in U if $I(\gamma; w) = 0$ for every $w \in \mathbb{C} \backslash U$.

Intuitively – γ does not wind around any of the 'holes' of U (points outside U). Note this is automatic if $U = \mathbb{C}$.

Before the main theorem, we need a somewhat technical lemma on difference quotients.

Lemma 5.6 (Difference Quotients are Jointly Continuous)
Let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be holomorphic. Define $g : U \times U \to \mathbb{C}$ by

$$g(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w. \end{cases}$$

Then g is continuous. Moreover, for any closed curve γ in U, the function $h(w) = \int_\gamma g(z, w)\, \mathrm{d}z$ is holomorphic on U.

Proof. Continuity of g is clear at points (z, w) with $z \neq w$. For a point (a, a), let $\varepsilon > 0$, and use continuity of f' (recall f' is holomorphic, in particular continuous) to pick $\delta > 0$ with $D(a, \delta) \subseteq U$ and $|f'(\zeta) - f'(a)| < \varepsilon$ for $\zeta \in D(a, \delta)$. For $z, w \in D(a, \delta)$ with $z \neq w$, the segment from w to z lies in $D(a, \delta)$, and

$$f(z) - f(w) = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} f(tz + (1-t)w)\, \mathrm{d}t = (z - w) \int_0^1 f'(tz + (1-t)w)\, \mathrm{d}t,$$

so that

$$\begin{aligned} |g(z, w) - g(a, a)| &= \left| \int_0^1 \left[f'(tz + (1-t)w) - f'(a) \right] \mathrm{d}t \right| \\ &\leq \sup_{t \in [0,1]} |f'(tz + (1-t)w) - f'(a)| < \varepsilon. \end{aligned}$$

The case $z = w$ is simply $|f'(z) - f'(a)| < \varepsilon$. Hence g is continuous at (a, a).

Now let γ be a closed curve in U and $h(w) = \int_\gamma g(z, w)\, \mathrm{d}z$. First, h is continuous: if $w_n \to w_0$ in U, choose $\delta > 0$ with $\overline{D(w_0, \delta)} \subset U$; then g is uniformly continuous on the compact set $\text{image}(\gamma) \times \overline{D(w_0, \delta)}$, so $g(\cdot, w_n) \to g(\cdot, w_0)$ uniformly on $\text{image}(\gamma)$, and hence $h(w_n) \to h(w_0)$ by our uniform convergence lemma.

To see h is holomorphic, we use Morera. Fix $w_0 \in U$ and a disk $D(w_0, \delta) \subseteq U$, and let β be any closed curve in $D(w_0, \delta)$. Writing out both integrals as integrals over parameter intervals, Fubini's theorem[a] lets us swap the order of integration:

$$\int_\beta h(w)\, \mathrm{d}w = \int_\beta \left(\int_\gamma g(z, w)\, \mathrm{d}z \right) \mathrm{d}w = \int_\gamma \left(\int_\beta g(z, w)\, \mathrm{d}w \right) \mathrm{d}z.$$

But for each fixed z, the function $w \mapsto g(z, w)$ is continuous on U and holomorphic on $U \setminus \{z\}$, hence holomorphic on U by our corollary on removing exceptional points. Since $D(w_0, \delta)$ is convex, $\int_\beta g(z, w)\, \mathrm{d}w = 0$ by Cauchy's theorem. Hence $\int_\beta h(w)\, \mathrm{d}w = 0$ for all closed β in $D(w_0, \delta)$, so h is holomorphic on $D(w_0, \delta)$ by Morera's theorem, and w_0 was arbitrary. $\square$

[a]For continuous integrands on products of compact intervals, Fubini's theorem – that the two iterated integrals agree – follows by uniform continuity: approximate the integrand uniformly by step functions, for which the interchange is immediate.

Theorem 5.7 (General Cauchy Theorem & Integral Formula)

Let $U \subseteq \mathbb{C}$ be a non-empty open set and γ a closed curve in U, homologous to zero in U. Then for every holomorphic $f : U \to \mathbb{C}$:

(i) $I(\gamma; w) f(w) = \dfrac{1}{2\pi i} \displaystyle\int_\gamma \frac{f(z)}{z - w}\, \mathrm{d}z$ for every $w \in U \setminus \text{image}(\gamma)$;

(ii) $\displaystyle\int_\gamma f(z)\, \mathrm{d}z = 0.$

Proof. Let us first note that (i) implies (ii): apply (i) to the holomorphic function $F(z) = (z - w) f(z)$ for any fixed $w \in U \setminus \text{image}(\gamma)$; since $F(w) = 0$, we get $\int_\gamma f(z)\, \mathrm{d}z = 0$.

So we prove (i), which (multiplying up) is precisely the statement that $h(w) = \int_\gamma g(z, w)\, \mathrm{d}z = 0$ for all $w \in U \setminus \text{image}(\gamma)$, where g is the difference quotient function of the previous lemma. By that lemma, h is holomorphic on U. The plan is to extend h to an entire function which vanishes at infinity, and then invoke Liouville.

Let

$$V = \{w \in \mathbb{C} \setminus \text{image}(\gamma) \mid I(\gamma; w) = 0\}.$$

Since γ is homologous to zero in U, we have $\mathbb{C} \setminus U \subseteq V$, i.e. $U \cup V = \mathbb{C}$. Also V is open, since $I(\gamma; \cdot)$ is locally constant. On the overlap $U \cap V$, for $w \in U \cap V$,

$$h(w) = \int_\gamma \frac{f(z) - f(w)}{z - w}\, \mathrm{d}z = \int_\gamma \frac{f(z)}{z - w}\, \mathrm{d}z - 2\pi i\, I(\gamma; w) f(w) = \int_\gamma \frac{f(z)}{z - w}\, \mathrm{d}z,$$

as $I(\gamma; w) = 0$ there. So h agrees on $U \cap V$ with the function

$$h_1(w) = \int_\gamma \frac{f(z)}{z - w}\, \mathrm{d}z, \quad w \in V,$$

which is holomorphic on V (by the standard argument: it is continuous, and holomorphic under the integral sign – or simply repeat the Morera argument of the lemma). Hence

$$H(w) = \begin{cases} h(w) & \text{if } w \in U, \\ h_1(w) & \text{if } w \in V \end{cases}$$

is a well-defined entire function.

Now pick R with $\text{image}(\gamma) \subset D(0,R)$. Then $\mathbb{C} \backslash D(0,R)$ lies in the unbounded component of the complement of γ, where the winding number vanishes, so $\mathbb{C} \backslash D(0,R) \subseteq V$. For $|w| > R$,

$$|H(w)| = |h_1(w)| \leq \frac{\text{length}(\gamma)}{|w| - R} \sup_{z \in \text{image}(\gamma)} |f(z)| \to 0 \quad \text{as } |w| \to \infty.$$

In particular H is bounded (continuous on a compact disk, small outside it), so by Liouville's theorem H is constant – and since $H(w) \to 0$, that constant is 0. Hence $h \equiv 0$ on U, as required. □

This machine has some immediate and very useful corollaries about systems of curves.

Corollary 5.8 (Cauchy's Theorem for Several Curves)
Let $U \subseteq \mathbb{C}$ be open and let $\gamma_1, \ldots, \gamma_n$ be closed curves in U such that

$$\sum_{j=1}^{n} I(\gamma_j; w) = 0 \quad \text{for all } w \in \mathbb{C} \backslash U.$$

Then for every holomorphic $f : U \to \mathbb{C}$,

$$\sum_{j=1}^{n} \int_{\gamma_j} f(z) \, \mathrm{d}z = 0, \quad \text{and} \quad f(w) \sum_{j=1}^{n} I(\gamma_j; w) = \sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\gamma_j} \frac{f(z)}{z - w} \, \mathrm{d}z$$

for every $w \in U$ not on any of the curves.

Proof. Run the proof of the theorem again, defining $h(w) = \sum_j \int_{\gamma_j} g(z,w) \, \mathrm{d}z$ and $V = \{w \mid \sum_j I(\gamma_j; w) = 0\}$; every step goes through verbatim. □

Corollary 5.9 (Curves with Equal Winding Numbers)
Let $U \subseteq \mathbb{C}$ be open and β_1, β_2 be closed curves in U with $I(\beta_1; w) = I(\beta_2; w)$ for every $w \in \mathbb{C} \backslash U$. Then

$$\int_{\beta_1} f(z) \, \mathrm{d}z = \int_{\beta_2} f(z) \, \mathrm{d}z$$

for every holomorphic $f : U \to \mathbb{C}$.

Proof. Apply the previous corollary to the curves $\gamma_1 = \beta_1$ and $\gamma_2 = -\beta_2$, noting $I(-\beta_2; w) = -I(\beta_2; w)$. □

§5.3 Homotopy and Simply Connected Domains

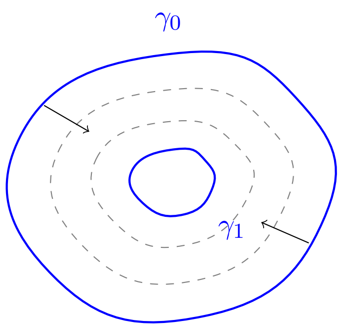
The condition 'homologous to zero' is exactly the right one, but it can be awkward to verify directly. A more intuitive (and slightly stronger) condition comes from the notion of *continuously deforming* curves.

Definition 5.10 (Homotopy)
Let $U \subseteq \mathbb{C}$ be a domain and $\gamma_0, \gamma_1 : [a, b] \to U$ be closed curves. We say γ_0 and γ_1 are **homotopic** in U if there is a continuous map $H : [0, 1] \times [a, b] \to U$ such that

$$H(0, t) = \gamma_0(t), \quad H(1, t) = \gamma_1(t), \quad H(s, a) = H(s, b)$$

for all $s \in [0, 1]$ and $t \in [a, b]$.

Writing $\gamma_s(t) = H(s, t)$, the definition says exactly that $\{\gamma_s\}_{s\in[0,1]}$ is a family of closed curves in U deforming γ_0 continuously into γ_1, without ever leaving U.



Definition 5.11 (Null-Homotopic)
A closed curve in a domain U is **null-homotopic** if it is homotopic in U to a constant curve.

The key fact is that deforming a curve cannot change any of the winding numbers that matter.

Theorem 5.12 (Homotopic Curves have Equal Winding Numbers)
If $\gamma_0, \gamma_1 : [a, b] \to U$ are homotopic closed curves in a domain U, then $I(\gamma_0; w) = I(\gamma_1; w)$ for every $w \in \mathbb{C}\backslash U$. In particular, a null-homotopic closed curve is homologous to zero in U.

Proof. We will use the following fact, which is an exercise on the example sheets: if $\gamma, \tilde{\gamma}$ are closed curves with

$$|\gamma(t) - \tilde{\gamma}(t)| < |w - \gamma(t)| \quad \text{for all } t,$$

then $I(\gamma; w) = I(\tilde{\gamma}; w)$. (Intuitively – if $\tilde{\gamma}$ stays closer to γ than γ is to w, then in walking along $\tilde{\gamma}$ instead of γ we can never gain or lose a loop around w.)

Let H be a homotopy from γ_0 to γ_1, and fix $w \in \mathbb{C}\backslash U$. The image $K = H([0, 1] \times [a, b])$ is compact and contained in U, so there is $\varepsilon > 0$ with $|w - H(s, t)| > 2\varepsilon$ for all (s,t). By uniform continuity of H, there is $n \in \mathbb{N}$ such that

$$|s-s'| + |t-t'| \leq \frac{\max\{1, b-a\}}{n} \implies |H(s,t) - H(s',t')| < \varepsilon.$$

For $k = 0, 1, \ldots, n$ let $\Gamma_k(t) = H(k/n, t)$, closed curves interpolating from γ_0 to γ_1 with $|\Gamma_{k-1}(t) - \Gamma_k(t)| < \varepsilon < |w - \Gamma_{k-1}(t)|$ for all t.

If each Γ_k were piecewise C^1 we would be done by the example sheet fact. In general they are merely continuous, so we approximate: let $a_j = a + j(b-a)/n$ for $0 \leq j \leq n$, and let $\tilde{\Gamma}_k$ be the closed polygonal curve agreeing with Γ_k at each a_j and affine on each $[a_{j-1}, a_j]$. On $[a_{j-1}, a_j]$, the value $\tilde{\Gamma}_k(t)$ is a convex combination of $\Gamma_k(a_{j-1})$ and $\Gamma_k(a_j)$, both within ε of $\Gamma_k(t)$ by uniform continuity; so $|\tilde{\Gamma}_k(t) - \Gamma_k(t)| < \varepsilon$ for all t. Next we estimate the distance between consecutive polygonal curves: $\tilde{\Gamma}_{k-1}(t) - \tilde{\Gamma}_k(t)$ is a convex combination of $\Gamma_{k-1}(a_{j-1}) - \Gamma_k(a_{j-1})$ and $\Gamma_{k-1}(a_j) - \Gamma_k(a_j)$, each of modulus less than ε, so $|\tilde{\Gamma}_{k-1}(t) - \tilde{\Gamma}_k(t)| < \varepsilon$. Also

$$|w - \tilde{\Gamma}_{k-1}(t)| \geq |w - \Gamma_{k-1}(t)| - |\Gamma_{k-1}(t) - \tilde{\Gamma}_{k-1}(t)| > 2\varepsilon - \varepsilon = \varepsilon.$$

So the example sheet fact applies to each consecutive pair, giving $I(\tilde{\Gamma}_0; w) = I(\tilde{\Gamma}_1; w) = \cdots = I(\tilde{\Gamma}_n; w)$. Finally, $|\tilde{\Gamma}_0(t) - \gamma_0(t)| < \varepsilon < |w - \gamma_0(t)|$ gives $I(\tilde{\Gamma}_0; w) = I(\gamma_0; w)$, and similarly at the other end, $I(\tilde{\Gamma}_n; w) = I(\gamma_1; w)$. Hence $I(\gamma_0; w) = I(\gamma_1; w)$.

For the last statement: a constant curve has all winding numbers zero, so a null-homotopic curve has $I(\gamma; w) = 0$ for every $w \notin U$. ☐

Corollary 5.13 (Deformation Invariance of Integrals)
If γ_0, γ_1 are homotopic closed curves in a domain U, then

$$\int_{\gamma_0} f(z)\, \mathrm{d}z = \int_{\gamma_1} f(z)\, \mathrm{d}z$$

for every holomorphic $f : U \to \mathbb{C}$.

Proof. Combine the theorem with our corollary on curves with equal winding numbers. ☐

Definition 5.14 (Simply Connected)
A domain U is **simply connected** if every closed curve in U is null-homotopic.

Intuitively, simply connected domains are the domains 'without holes'. For example, every star-shaped domain U with centre a_0 is simply connected: given a closed curve γ, the map $H(s,t) = (1-s)\gamma(t) + s a_0$ is a homotopy in U (each point $H(s,t)$ lies on the segment from $\gamma(t)$ to the centre) from γ to the constant curve at a_0.

Theorem 5.15 (Cauchy's Theorem for Simply Connected Domains)

Let U be a simply connected domain and $f : U \to \mathbb{C}$ be holomorphic. Then

$$\int_\gamma f(z)\,\mathrm{d}z = 0$$

for every closed curve γ in U.

Proof. Every closed curve in U is null-homotopic, hence homologous to zero, so the general Cauchy theorem applies. □

Remark. The converse is also true (though harder to prove): if $\int_\gamma f\ \mathrm{d}z = 0$ for all holomorphic f and closed curves γ in U, then U is simply connected. However, at the level of individual curves the two notions genuinely differ: in $U = \mathbb{C}\backslash\{w_1, w_2\}$ (two punctures), there are closed curves that are homologous to zero but *not* null-homotopic – the classic example traces around both punctures in a clever 'figure-of-eight-like' pattern so that both winding numbers cancel to zero, yet the curve cannot be shrunk to a point. Homology counts winding; homotopy remembers more.

§6 Isolated Singularities and Laurent Series

The Cauchy integral formula lets us integrate functions of the form $f(z)/(z-a)$ where f is holomorphic at a. But many natural functions – say $e^{1/z}$ near $z = 0$ – fail to be holomorphic at a point in a more serious way, and we would still like to integrate them around that point. This leads us to classify how a holomorphic function can behave near an isolated 'bad point', and to a generalisation of Taylor series adapted to such points.

§6.1 Removable Singularities and Poles

Definition 6.1 (Isolated Singularity)

Let $U \subseteq \mathbb{C}$ be open and $a \in U$. If $f : U\backslash\{a\} \to \mathbb{C}$ is holomorphic, we say f has an **isolated singularity** at a.

The mildest possibility is that the singularity is illusory.

Definition 6.2 (Removable Singularity)

An isolated singularity a of f is a **removable singularity** if f can be defined at a in such a way that the extended function is holomorphic on U.

Proposition 6.3 (Characterising Removable Singularities)

Let U be open, $a \in U$, and $f : U\backslash\{a\} \to \mathbb{C}$ be holomorphic. The following are equivalent:

(i) f has a removable singularity at a;

(ii) $\lim_{z\to a} f(z)$ exists in $\mathbb{C}$;

(iii) $|f|$ is bounded on $D(a,\varepsilon)\backslash\{a\}$ for some $\varepsilon > 0$;

(iv) $\lim_{z\to a}(z-a)f(z) = 0$.

Proof. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) are each immediate. The content is (iv) $\Rightarrow$ (i). Define $h : U \to \mathbb{C}$ by

$$h(z) = \begin{cases} (z-a)^2 f(z) & \text{if } z \neq a, \\ 0 & \text{if } z = a. \end{cases}$$

Then h is holomorphic on $U \setminus \{a\}$, and at a,

$$\lim_{z\to a} \frac{h(z) - h(a)}{z-a} = \lim_{z\to a}(z-a)f(z) = 0,$$

using (iv). So h is differentiable at a too, with $h(a) = h'(a) = 0$, i.e. h is holomorphic on U. Its Taylor expansion at a therefore starts at the quadratic term: on some $D(a,r)$ we have $h(z) = (z-a)^2 g(z)$ with g holomorphic on $D(a,r)$. Comparing with the definition of h, we get $f = g$ on $D(a,r)\setminus\{a\}$, so defining $f(a) = g(a)$ extends f holomorphically. $\square$

Example 6.4

The function $f(z) = \frac{e^z-1}{z}$ is holomorphic on $\mathbb{C}\setminus\{0\}$, and $zf(z) = e^z - 1 \to 0$ as $z \to 0$, so the singularity at 0 is removable. Indeed, dividing the Taylor series of $e^z - 1$ by z exhibits the extension $\sum_{k\geq 1} z^{k-1}/k!$ directly, an entire function.

If a singularity is not removable, then by (iii) above f is unbounded near a. The tamest way this can happen is if f blows up cleanly.

Definition 6.5 (Pole)

An isolated singularity a of f is a **pole** if $|f(z)| \to \infty$ as $z \to a$.

Definition 6.6 (Essential Singularity)

An isolated singularity which is neither removable nor a pole is called an **essential singularity**.

Example 6.7

(i) $f(z) = (z-a)^{-k}$ for $k \in \mathbb{N}$ has a pole at a.

(ii) $f(z) = e^{1/z}$ has an essential singularity at 0. Indeed, $|f(iy)| = 1$ for real $y \neq 0$, while $f(x) \to \infty$ as $x \to 0^+$ along the reals – so $|f|$ neither stays bounded nor tends to infinity.

Poles are completely understood in terms of zeros of holomorphic functions.

Proposition 6.8 (Characterising Poles)

Let $f : U\setminus\{a\} \to \mathbb{C}$ be holomorphic. The following are equivalent:

(i) f has a pole at a;

(ii) there is $\varepsilon > 0$ and a holomorphic $h : D(a,\varepsilon) \to \mathbb{C}$ with $h(a) = 0$, $h \neq 0$ on $D(a,\varepsilon)\backslash\{a\}$, and $f = 1/h$ on $D(a,\varepsilon)\backslash\{a\}$;

(iii) there is a unique integer $k \geq 1$ and a unique holomorphic $g : U \to \mathbb{C}$ with $g(a) \neq 0$ such that

$$f(z) = (z-a)^{-k} g(z) \quad \text{on } U\backslash\{a\}.$$

Proof. *(i) ⇒ (ii).* Since $|f(z)| \to \infty$, there is $\varepsilon > 0$ with $|f| \geq 1$ on $D(a,\varepsilon)\backslash\{a\}$, so $1/f$ is holomorphic and bounded (by 1) there. By the removable singularity criterion, $1/f$ extends to a holomorphic h on $D(a,\varepsilon)$, and since $|f| \to \infty$ we must have $h(a) = 0$; clearly $h \neq 0$ away from a and $f = 1/h$.

(ii) ⇒ (iii). By our factorisation of zeros, $h(z) = (z-a)^k h_1(z)$ for some $k \geq 1$ and holomorphic h_1 with $h_1 \neq 0$ on $D(a,\varepsilon)$ (shrinking ε if needed). Then $g_1 = 1/h_1$ is holomorphic on $D(a,\varepsilon)$ and $f(z) = (z-a)^{-k} g_1(z)$ there. Now define $g : U \to \mathbb{C}$ by $g = g_1$ on $D(a,\varepsilon)$ and $g(z) = (z-a)^k f(z)$ on $U\backslash\{a\}$; the two definitions agree on the overlap, so g is well-defined, holomorphic, and $g(a) = g_1(a) \neq 0$.

For uniqueness, if $(z-a)^{-k} g(z) = (z-a)^{-\tilde{k}} \tilde{g}(z)$ with $g(a), \tilde{g}(a) \neq 0$, then $g(z) = (z-a)^{k-\tilde{k}} \tilde{g}(z)$ away from a; if $k \neq \tilde{k}$ this forces one of $g(a), \tilde{g}(a)$ to vanish (letting $z \to a$), a contradiction. So $k = \tilde{k}$, and then $g = \tilde{g}$ off a, hence everywhere by continuity.

(iii) ⇒ (i). Immediate: $|f(z)| = |z-a|^{-k} |g(z)| \to \infty$ as $z \to a$ since $g(a) \neq 0$. □

Definition 6.9 (Order of a Pole)

If f has a pole at a, the unique integer $k \geq 1$ above is the **order** of the pole. A pole of order 1 is called a **simple pole**.

Definition 6.10 (Meromorphic)

Let U be open and $S \subset U$ be a discrete subset (all of its points are isolated). If $f : U\backslash S \to \mathbb{C}$ is holomorphic and every point of S is a removable singularity or a pole of f, we say f is **meromorphic** on U.

Remark (The Riemann Sphere)**.** A pole is not really a ‘bad point’ at all, if one is willing to enlarge the target. Adding a point at infinity to $\mathbb{C}$ yields the *Riemann sphere* $\mathbb{C} \cup \{\infty\}$, and a meromorphic function on U becomes a genuinely continuous (indeed holomorphic, suitably interpreted) map $U \to \mathbb{C} \cup \{\infty\}$ by sending poles to ∞. From this point of view the only true singularities are essential ones. This perspective is developed properly in the Part II course *Riemann Surfaces*.

Near an essential singularity, on the other hand, the behaviour is truly wild.

Theorem 6.11 (Casorati-Weierstrass)

Let $f : U\backslash\{a\} \to \mathbb{C}$ be holomorphic with an essential singularity at a. Then for every $\varepsilon > 0$, the image $f(D(a,\varepsilon)\backslash\{a\})$ is *dense* in $\mathbb{C}$: for any $b \in \mathbb{C}$ and any $\delta > 0$, there are points z arbitrarily close to a with $|f(z) - b| < \delta$.

Proof. Suppose not: there are $b \in \mathbb{C}$, $\delta > 0$ and $\varepsilon > 0$ such that $|f(z) - b| \geq \delta$ for all $z \in D(a, \varepsilon) \setminus \{a\}$. Then

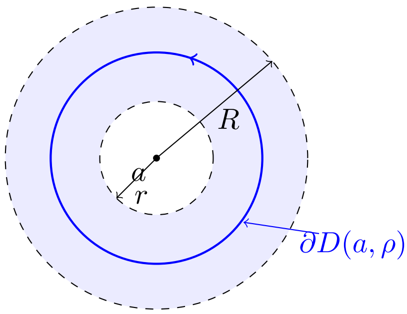
$$g(z) = \frac{1}{f(z) - b}$$

is holomorphic and bounded (by $1/\delta$) on $D(a, \varepsilon)\setminus\{a\}$, so g extends holomorphically over a. If $g(a) \neq 0$ then $f = b + 1/g$ is bounded near a, so a is removable; if $g(a) = 0$ then $|f(z)| = |b + 1/g(z)| \to \infty$ as $z \to a$, so a is a pole. Either way we contradict the singularity being essential. □

Remark (Picard's Theorem)**.** Much more is true: *Picard's theorem* states that near an essential singularity, f attains every complex value, with at most a single exception, infinitely often. (For $e^{1/z}$ the exceptional value is 0.) The proof is well beyond this course.

§6.2 Laurent Series

We can now generalise Taylor series to functions holomorphic on an *annulus* – in particular on a punctured disk, the natural home of an isolated singularity. The price is that negative powers of $(z - a)$ appear; the reward is that they tell us everything about the singularity.



Theorem 6.12 (Laurent Expansion)
Let $0 \leq r < R \leq \infty$ and let f be holomorphic on the annulus

$$A = \{z \in \mathbb{C} \mid r < |z - a| < R\}.$$

Then:

(i) f has a unique convergent series expansion on A,

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n = \sum_{n=1}^{\infty} c_{-n}(z - a)^{-n} + \sum_{n=0}^{\infty} c_n (z - a)^n;$$

(ii) for any $\rho \in (r, R)$, the coefficients are given by

$$c_n = \frac{1}{2\pi i} \int_{\partial D(a,\rho)} \frac{f(z)}{(z - a)^{n+1}} \, \mathrm{d}z;$$

(iii) if $r < \rho' \le \rho < R$, the two series in (i) converge uniformly on $\{\rho' \le |z-a| \le \rho\}$.

Proof. Fix $w \in A$, and choose ρ_1, ρ_2 with $r < \rho_1 < |w-a| < \rho_2 < R$. As in the proof of the Cauchy integral formula, the function

$$g(z) = \begin{cases} \frac{f(z)-f(w)}{z-w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w \end{cases}$$

is continuous on A and holomorphic on $A \setminus \{w\}$, hence holomorphic on A (removable singularity). The circles $\partial D(a, \rho_1)$ and $\partial D(a, \rho_2)$ are homotopic in A (deform the radius continuously), so

$$\int_{\partial D(a,\rho_1)} g(z)\, \mathrm{d}z = \int_{\partial D(a,\rho_2)} g(z)\, \mathrm{d}z.$$

Substituting the definition of g and using $I(\partial D(a,\rho_1); w) = 0$ (as $|w-a| > \rho_1$) and $I(\partial D(a,\rho_2); w) = 1$, we obtain the annular Cauchy integral formula

$$f(w) = \frac{1}{2\pi i} \int_{\partial D(a,\rho_2)} \frac{f(z)}{z-w}\, \mathrm{d}z - \frac{1}{2\pi i} \int_{\partial D(a,\rho_1)} \frac{f(z)}{z-w}\, \mathrm{d}z.$$

Now expand each integrand in a geometric series, exactly as for Taylor series. On $\partial D(a, \rho_2)$ we have $|w-a| < |z-a|$, so

$$\frac{1}{z-w} = \sum_{n=0}^{\infty} \frac{(w-a)^n}{(z-a)^{n+1}},$$

uniformly in z; while on $\partial D(a, \rho_1)$ we have $|z-a| < |w-a|$, so

$$\frac{1}{z-w} = -\sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}},$$

again uniformly. Integrating term by term (using our uniform convergence lemma) gives

$$f(w) = \sum_{n=0}^{\infty} c_n (w-a)^n + \sum_{n=1}^{\infty} c_{-n} (w-a)^{-n},$$

where c_n (for $n \ge 0$) is given by the integral over $\partial D(a, \rho_2)$ as in (ii), and c_{-n} (for $n \ge 1$) by the integral over $\partial D(a, \rho_1)$. Since any two circles $\partial D(a, \rho)$, $\rho \in (r, R)$, are homotopic in A and the integrands are holomorphic on A, the integrals do not depend on the radius used, giving (ii) as stated.

For (iii) and uniqueness, suppose $f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$ on A for some coefficients c_n. The non-negative part $\sum_{n \ge 0} c_n (z-a)^n$ converges on A, so its radius of convergence is at least R, giving uniform convergence on $|z-a| \le \rho < R$. For the negative part, substitute $\zeta = (z-a)^{-1}$: the series $\sum_{n \ge 1} c_{-n} \zeta^n$ converges for $1/R < |\zeta| < 1/r$, so has radius of convergence at least $1/r$, and converges uniformly on $|\zeta| \le 1/\rho'$, i.e. on $|z-a| \ge \rho'$. This proves (iii). Then for any $m \in \mathbb{Z}$ we may integrate the uniformly convergent series term by term against $(z-a)^{-m-1}$:

$$\int_{\partial D(a,\rho)} \frac{f(z)}{(z-a)^{m+1}}\,\mathrm{d}z = \sum_{n=-\infty}^{\infty} c_n \int_{\partial D(a,\rho)} (z-a)^{n-m-1}\,\mathrm{d}z = 2\pi i\, c_m,$$

since $\int_{\partial D(a,\rho)} (z-a)^j\,\mathrm{d}z = 0$ for $j \ne -1$ (antiderivative!) and $= 2\pi i$ for $j=-1$. So the coefficients are forced to be those of (ii), giving uniqueness. □

Remark. If f happens to extend holomorphically to the full disk $D(a,R)$, then each coefficient c_{-m} with $m \ge 1$ vanishes by Cauchy's theorem (the integrand is then holomorphic on the disk), and the Laurent series is just the Taylor series. The new content is precisely for functions with no such extension.

Applying this on a punctured disk ($r=0$), the shape of the Laurent series reads off the type of singularity.

Proposition 6.13 (Classification of Singularities via Laurent Series)
Let f be holomorphic on the punctured disk $D(a,R)\setminus\{a\}$, with Laurent expansion $f(z) = \sum_{n\in\mathbb{Z}} c_n (z-a)^n$. Then:

(i) a is a removable singularity if and only if $c_n = 0$ for all $n < 0$;

(ii) a is a pole of order k if and only if $c_{-k} \ne 0$ and $c_n = 0$ for all $n < -k$;

(iii) a is an essential singularity if and only if $c_n \ne 0$ for infinitely many $n < 0$.

Proof. If $c_n = 0$ for $n < 0$, the series defines a holomorphic function on $D(a,R)$ agreeing with f away from a, so the singularity is removable; conversely if the singularity is removable the Laurent series is the Taylor series of the extension.

If $c_{-k} \ne 0$ and lower coefficients vanish, then $f(z) = (z-a)^{-k} g(z)$ where $g(z) = \sum_{n\ge 0} c_{n-k}(z-a)^n$ is holomorphic on $D(a,R)$ with $g(a) = c_{-k} \ne 0$; by our characterisation, a is a pole of order k. Conversely a pole of order k has this form, and expanding g in a Taylor series produces such a Laurent series (unique!).

Finally (iii) holds since (i) and (ii) are equivalences and the three cases are exhaustive and mutually exclusive. □

Example 6.14 (Laurent Series of $e^{1/z}$)
Substituting $1/z$ into the exponential series,

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{z^{-n}}{n!} = 1 + \frac{1}{z} + \frac{1}{2!\,z^2} + \cdots, \quad z \ne 0.$$

Infinitely many negative coefficients are non-zero, confirming that 0 is an essential singularity.

Example 6.15 (Different Annuli, Different Series)

A function can have different Laurent expansions on different annuli. Take

$$f(z) = \frac{1}{z(1-z)} = \frac{1}{z} + \frac{1}{1-z}.$$

On the punctured disk $0 < |z| < 1$ we expand $\frac{1}{1-z}$ as a geometric series:

$$f(z) = \frac{1}{z} + 1 + z + z^2 + \cdots,$$

exhibiting a simple pole at 0. But on the annulus $1 < |z| < \infty$ we must instead expand in powers of $1/z$:

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-1/z} = -\frac{1}{z} - \frac{1}{z^2} - \cdots,$$

so there $f(z) = -z^{-2} - z^{-3} - \cdots$. The moral: a Laurent expansion is always relative to a choice of annulus.

§7 The Residue Theorem

We now assemble everything into the single most useful computational theorem of the course. It turns out that when integrating around an isolated singularity, only *one* Laurent coefficient survives.

§7.1 Residues

Definition 7.1 (Residue and Principal Part)

Let $f : D(a,R)\backslash\{a\} \to \mathbb{C}$ be holomorphic, with Laurent expansion $\sum_{n=-\infty}^{\infty} c_n(z-a)^n$. The **residue** of f at a is

$$\operatorname{Res}_{z=a} f = c_{-1},$$

and the **principal part** of f at a is the series

$$f_P(z) = \sum_{n=1}^{\infty} c_{-n}(z-a)^{-n}.$$

Why is c_{-1} singled out? Because $(z-a)^n$ has an antiderivative for every n *except* $n = -1$. So when we integrate the Laurent series term by term around a closed curve, only the c_{-1} term can contribute – and it contributes exactly $2\pi i\, c_{-1}$ times the winding number. Let us make this precise, in full generality.

Note first that the series defining f_P converges for all $z \neq a$ (its convergence in $0 < |z-a| < R$ forces, via the substitution $\zeta = (z-a)^{-1}$, a radius of convergence of at least $1/r$ for every $r > 0$), and the convergence is uniform on compact subsets of $\mathbb{C}\backslash\{a\}$. In particular f_P is holomorphic on $\mathbb{C}\backslash\{a\}$, and $f - f_P$ has a removable singularity at a.

Theorem 7.2 (Residue Theorem)

Let $U \subseteq \mathbb{C}$ be open, let $a_1, \ldots, a_k \in U$ be distinct points, and let

$$f : U \backslash \{a_1, \ldots, a_k\} \to \mathbb{C}$$

be holomorphic. If γ is a closed curve in U, homologous to zero in U, with no a_j on the curve, then

$$\frac{1}{2\pi i} \int_\gamma f(z)\, \mathrm{d}z = \sum_{j=1}^{k} I(\gamma; a_j) \operatorname{Res}_{z=a_j} f.$$

Proof. Let $f_P^{(j)}$ be the principal part of f at a_j, a holomorphic function on $\mathbb{C}\backslash\{a_j\}$, and set

$$h = f - \left(f_P^{(1)} + \cdots + f_P^{(k)}\right),$$

holomorphic on $U\backslash\{a_1, \ldots, a_k\}$. Fix j: near a_j, the difference $f - f_P^{(j)}$ has a removable singularity (its Laurent series has no negative terms), and each $f_P^{(\ell)}$ with $\ell \neq j$ is holomorphic at a_j. So h has a removable singularity at each a_j, and extends to a holomorphic function on all of U. By the general Cauchy theorem, $\int_\gamma h(z)\, \mathrm{d}z = 0$, so

$$\frac{1}{2\pi i} \int_\gamma f(z)\, \mathrm{d}z = \sum_{j=1}^{k} \frac{1}{2\pi i} \int_\gamma f_P^{(j)}(z)\, \mathrm{d}z.$$

Finally, the series for $f_P^{(j)}$ converges uniformly on the (compact) image of γ, so we may integrate term by term; every term $(z - a_j)^{-n}$ with $n \geq 2$ has an antiderivative on $\mathbb{C}\backslash\{a_j\}$ and integrates to zero, leaving

$$\frac{1}{2\pi i} \int_\gamma f_P^{(j)}(z)\, \mathrm{d}z = \frac{c_{-1}^{(j)}}{2\pi i} \int_\gamma \frac{\mathrm{d}z}{z - a_j} = I(\gamma; a_j) \operatorname{Res}_{z=a_j} f.$$

□

In practice γ will wind once around each singularity we care about and zero times around everything else, and then the theorem reads: *the integral is $2\pi i$ times the sum of the residues inside.*

§7.2 Computing Residues

To use the residue theorem we need to actually compute residues. Thankfully, for poles this rarely requires finding the whole Laurent series.

Proposition 7.3 (Residue Computation Rules)

(i) If f has a simple pole at a, then

$$\operatorname{Res}_{z=a} f = \lim_{z \to a} (z - a) f(z).$$

(ii) If $f(z) = (z - a)^{-k} g(z)$ with g holomorphic at a and $g(a) \neq 0$ (a pole of order

k), then

$$\operatorname{Res}_{z=a} f = \frac{g^{(k-1)}(a)}{(k-1)!}.$$

(iii) If $f = g/h$ with g, h holomorphic at a, $g(a) \neq 0$, and h having a simple zero at a, then

$$\operatorname{Res}_{z=a} f = \frac{g(a)}{h'(a)}.$$

Proof. For (ii) (which contains (i) as the case $k = 1$): expanding g in its Taylor series at a, the coefficient of $(z-a)^{-1}$ in the Laurent series of $(z-a)^{-k}g(z)$ is the coefficient of $(z-a)^{k-1}$ in the Taylor series of g, namely $g^{(k-1)}(a)/(k-1)!$.

For (iii): $h(z) = (z-a)h_1(z)$ with h_1 holomorphic, $h_1(a) = h'(a) \neq 0$, so f has a simple pole at a and by (i)

$$\operatorname{Res}_{z=a} f = \lim_{z \to a} \frac{(z-a)g(z)}{h(z)} = \lim_{z \to a} \frac{g(z)}{\left(\frac{h(z)-h(a)}{z-a}\right)} = \frac{g(a)}{h'(a)}.$$

□

Example 7.4

Let $f(z) = \frac{e^{iz}}{1+z^2}$. The denominator factors as $(z-i)(z+i)$, so f has simple poles at $\pm i$, and by rule (iii),

$$\operatorname{Res}_{z=i} f = \frac{e^{i \cdot i}}{2i} = \frac{e^{-1}}{2i}.$$

§7.3 Evaluating Real Integrals

We now come to the payoff: the residue theorem is a machine for computing real integrals that resist elementary methods. The general strategy is always the same:

(1) pick a holomorphic (or meromorphic) function related to the real integrand;

(2) pick a closed contour, part of which runs along (a piece of) the real axis;

(3) compute the contour integral by residues;

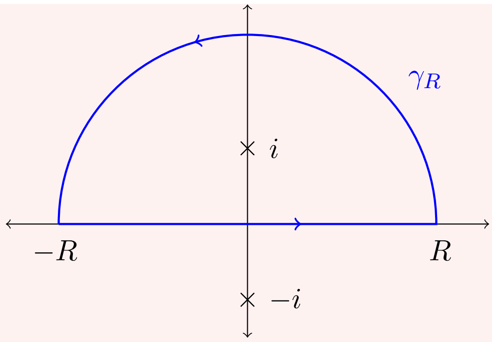
(4) show the contribution of the auxiliary parts of the contour vanishes (or converges to something known) in a limit.

Example 7.5 (A Semicircular Contour)

We compute

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1+x^2} \quad (= \pi, \text{ as we know from } \arctan).$$

Let $f(z) = \frac{1}{1+z^2}$, which has simple poles at $\pm i$. For $R > 1$, let γ be the closed contour consisting of the segment $[-R, R]$ along the real axis, followed by the semicircular arc $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$, in the upper half-plane.



The contour is a closed curve in the star-shaped domain $\mathbb{C}$ (so certainly homologous to zero), winding once around i and zero times around $-i$. By the residue theorem and rule (iii),

$$\int_\gamma f(z)\,\mathrm{d}z = 2\pi i \operatorname{Res}_{z=i} \frac{1}{1+z^2} = 2\pi i \cdot \frac{1}{2i} = \pi.$$

On the arc, $|1+z^2| \geq R^2 - 1$, so

$$\left| \int_{\gamma_R} f(z)\,\mathrm{d}z \right| \leq \pi R \cdot \frac{1}{R^2-1} \to 0 \quad \text{as } R \to \infty.$$

Hence

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1+x^2} = \lim_{R\to\infty} \int_{-R}^{R} \frac{\mathrm{d}x}{1+x^2} = \pi.$$

That example could have been done in first year – but the same contour computes many integrals that certainly could not. When the integrand involves oscillating factors like e^{ix}, the arc estimate needs more care, and the following lemma is the standard tool.

Lemma 7.6 (Jordan's Lemma)

Let f be a continuous complex-valued function on the semicircle $C_R^+ = \{Re^{it} \mid t \in [0,\pi]\}$, and let $\gamma_R^+(t) = Re^{it}$, $t \in [0,\pi]$. Then for $\alpha > 0$,

$$\left| \int_{\gamma_R^+} f(z) e^{i\alpha z}\,\mathrm{d}z \right| \leq \frac{\pi}{\alpha} \sup_{z \in C_R^+} |f(z)|.$$

In particular, if $\sup_{z\in C_R^+} |f(z)| \to 0$ as $R \to \infty$, then $\int_{\gamma_R^+} f(z) e^{i\alpha z}\,\mathrm{d}z \to 0$.

Proof. Let $M_R = \sup_{C_R^+} |f|$. On the arc, $|e^{i\alpha z}| = e^{-\alpha R \sin t}$, so

$$\left| \int_{\gamma_R^+} f(z) e^{i\alpha z}\,\mathrm{d}z \right| \leq \int_0^\pi M_R e^{-\alpha R \sin t} R\,\mathrm{d}t = 2RM_R \int_0^{\pi/2} e^{-\alpha R \sin t}\,\mathrm{d}t,$$

using the symmetry $\sin(\pi - t) = \sin t$. Now on $(0, \pi/2]$ we have the classical bound $\sin t \geq 2t/\pi$ (the function $\sin t/t$ decreases on this interval, with value $2/\pi$ at $\pi/2$), so

$$2RM_R \int_0^{\pi/2} e^{-\alpha R \sin t}\,\mathrm{d}t \leq 2RM_R \int_0^{\pi/2} e^{-2\alpha R t/\pi}\,\mathrm{d}t = \frac{\pi M_R}{\alpha}\left(1 - e^{-\alpha R}\right) \leq \frac{\pi M_R}{\alpha}.$$

□

The point of Jordan's lemma is that decay of f alone (however slow) suffices – the oscillation of $e^{i\alpha z}$ does the rest. A companion lemma handles small arcs around simple poles that sit *on* our contour.

Lemma 7.7 (Small Arcs around Simple Poles)
Let f be holomorphic on $D(a,R)\backslash\{a\}$ with a simple pole at a, and for $0<\varepsilon<R$ let $\gamma_\varepsilon : [\alpha,\beta] \to \mathbb{C}$ be the arc $\gamma_\varepsilon(t) = a + \varepsilon e^{it}$. Then

$$\lim_{\varepsilon\to 0^+} \int_{\gamma_\varepsilon} f(z)\, \mathrm{d}z = (\beta-\alpha)\, i \operatorname{Res}_{z=a} f.$$

Proof. Write $f(z) = \frac{c}{z-a} + g(z)$ with $c = \operatorname{Res}_{z=a} f$ and g holomorphic on $D(a,R)$ (this is exactly the Laurent expansion of a simple pole). Then g is bounded near a, say $|g| \le M$ on $D(a, R/2)$, so

$$\left| \int_{\gamma_\varepsilon} g(z)\, \mathrm{d}z \right| \le (\beta-\alpha)\varepsilon M \to 0,$$

while direct computation gives

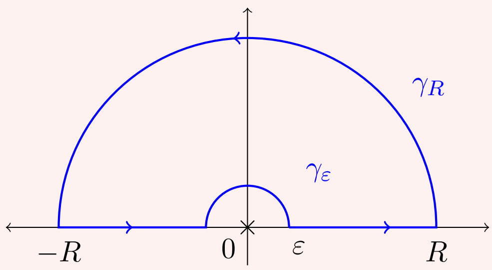
$$\int_{\gamma_\varepsilon} \frac{c}{z-a}\, \mathrm{d}z = c \int_\alpha^\beta \frac{i\varepsilon e^{it}}{\varepsilon e^{it}}\, \mathrm{d}t = (\beta-\alpha)ic.$$

□

Example 7.8 (The Dirichlet Integral)
We compute the famous integral

$$\int_0^\infty \frac{\sin x}{x}\, \mathrm{d}x = \frac{\pi}{2}.$$

The natural function to use is $f(z) = e^{iz}/z$, which has a simple pole at 0 – right on the contour we would like to use! So we *indent*: for $0<\varepsilon<R$, consider the closed contour γ consisting of the segment $[\varepsilon, R]$, the large semicircular arc $\gamma_R(t) = Re^{it}$, $t\in[0,\pi]$, the segment $[-R,-\varepsilon]$, and the small *clockwise* arc γ_ε from $-\varepsilon$ to ε over the top of the origin.



Now f is holomorphic on $\mathbb{C}\backslash\{0\}$, and γ winds zero times around 0 – the indentation was designed exactly to avoid it – so γ is homologous to zero in $\mathbb{C}\backslash\{0\}$, and the general Cauchy theorem gives $\int_\gamma f(z)\,\mathrm{d}z = 0$. That is,

$$\int_{[\varepsilon,R]} \frac{e^{ix}}{x}\,\mathrm{d}x + \int_{\gamma_R} f\,\mathrm{d}z + \int_{[-R,-\varepsilon]} \frac{e^{ix}}{x}\,\mathrm{d}x + \int_{\gamma_\varepsilon} f\,\mathrm{d}z = 0.$$

Substituting $x \mapsto -x$ in the third integral and combining with the first,

$$\int_\varepsilon^R \frac{e^{ix} - e^{-ix}}{x}\,\mathrm{d}x = 2i \int_\varepsilon^R \frac{\sin x}{x}\,\mathrm{d}x.$$

By Jordan's lemma (with $f(z) = 1/z$, $\sup_{C_R^+} |1/z| = 1/R \to 0$), the integral over γ_R tends to 0 as $R \to \infty$. The small arc γ_ε is traversed clockwise from angle π to 0, so by the small arcs lemma its integral tends to $-\pi i \operatorname{Res}_{z=0} f = -\pi i$ as $\varepsilon \to 0^+$. Putting everything together,

$$2i \int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x - \pi i = 0, \quad \text{that is} \quad \int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Next, an integral involving a branch cut, where the multi-valuedness of z^α is not an obstacle but the very thing that makes the method work.

Example 7.9 (A Keyhole Contour)

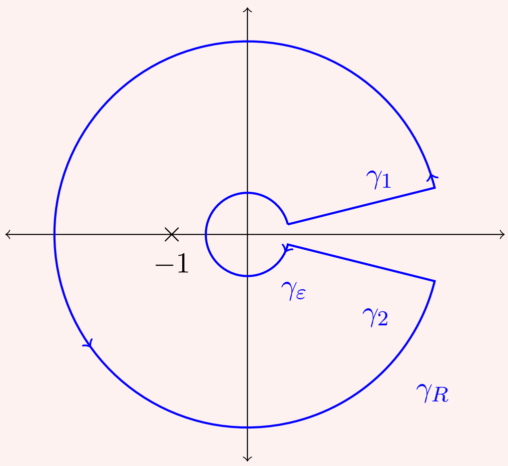
For $0 < \alpha < 1$, we show

$$\int_0^\infty \frac{x^{-\alpha}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin \pi\alpha}.$$

Work on the cut plane $U = \mathbb{C}\backslash\{x \in \mathbb{R} \mid x \geq 0\}$, on which we choose the branch $g(z) = z^{-\alpha} = e^{-\alpha \ell(z)}$, where $\ell(z) = \ln|z| + i\arg(z)$ with $\arg(z) \in (0, 2\pi)$. Let $f(z) = g(z)/(1+z)$, holomorphic on $U\backslash\{-1\}$ with a simple pole at -1 and

$$\operatorname{Res}_{z=-1} f = g(-1) = e^{-\alpha(\ln 1 + i\pi)} = e^{-i\pi\alpha}.$$

For $0 < \varepsilon < 1 < R$ and small $\theta > 0$, let γ be the 'keyhole contour' below: out along the ray $\arg z = \theta$ from radius ε to R, anticlockwise around the big circle to the ray $\arg z = 2\pi - \theta$, back along that ray, and clockwise around the small circle.



The domain U is not star-shaped, but the winding number of γ about every point of the positive real axis (and about 0) is zero, so γ is homologous to zero in U; and

$I(\gamma; -1) = 1$. By the residue theorem,

$$\int_\gamma f(z)\, \mathrm{d}z = 2\pi i\, e^{-i\pi\alpha}.$$

Consider the pieces. On the outgoing ray $\gamma_1(t) = te^{i\theta}$, $t \in [\varepsilon, R]$,

$$\int_{\gamma_1} f(z)\, \mathrm{d}z = \int_\varepsilon^R \frac{t^{-\alpha} e^{-i\alpha\theta} e^{i\theta}}{1 + te^{i\theta}}\, \mathrm{d}t \xrightarrow{\theta \to 0^+} \int_\varepsilon^R \frac{t^{-\alpha}}{1+t}\, \mathrm{d}t,$$

with uniform convergence of the integrands on $[\varepsilon, R]$. On the returning ray γ_2 ($\arg z = 2\pi - \theta$, traversed inwards from radius R down to radius ε), the branch of $z^{-\alpha}$ picks up an extra factor tending to $e^{-2\pi i\alpha}$ as $\theta \to 0^+$, so in that limit this piece contributes

$$-\, e^{-2\pi i\alpha} \int_\varepsilon^R \frac{t^{-\alpha}}{1+t}\, \mathrm{d}t,$$

the minus sign coming from the direction of traversal. Meanwhile, on the circular arcs we estimate

$$\left| \int_{\gamma_R} f\, \mathrm{d}z \right| \le 2\pi R \cdot \frac{R^{-\alpha}}{R-1} = \frac{2\pi R^{1-\alpha}}{R-1}, \qquad \left| \int_{\gamma_\varepsilon} f\, \mathrm{d}z \right| \le \frac{2\pi\varepsilon^{1-\alpha}}{1-\varepsilon},$$

both independent of θ. Since $0 < \alpha < 1$, both bounds vanish as $R \to \infty$, $\varepsilon \to 0^+$ respectively. Taking $\theta \to 0^+$, then $\varepsilon \to 0^+$ and $R \to \infty$,

$$\left(1 - e^{-2\pi i\alpha}\right) \int_0^\infty \frac{t^{-\alpha}}{1+t}\, \mathrm{d}t = 2\pi i e^{-i\pi\alpha},$$

and dividing through,

$$\int_0^\infty \frac{t^{-\alpha}}{1+t}\, \mathrm{d}t = \frac{2\pi i}{e^{i\pi\alpha} - e^{-i\pi\alpha}} = \frac{\pi}{\sin \pi\alpha}.$$

§7.4 Summing Series

The residue theorem can also sum infinite series. The trick is to find a meromorphic function with poles at all the integers with known residues – and $\pi \cot(\pi z)$, which has a simple pole of residue 1 at every integer, is precisely such a function.

Example 7.10 (The Basel Problem)

We prove

$$\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Consider

$$f(z) = \frac{\pi \cot(\pi z)}{z^2} = \frac{\pi \cos(\pi z)}{z^2 \sin(\pi z)}.$$

The zeros of $\sin(\pi z)$ are exactly the integers, and are simple. So f has simple poles at each $n \in \mathbb{Z}\backslash\{0\}$ with, by our rules,

$$\operatorname{Res}_{z=n} f = \frac{\pi \cos(\pi n)/n^2}{\pi \cos(\pi n)} = \frac{1}{n^2},$$

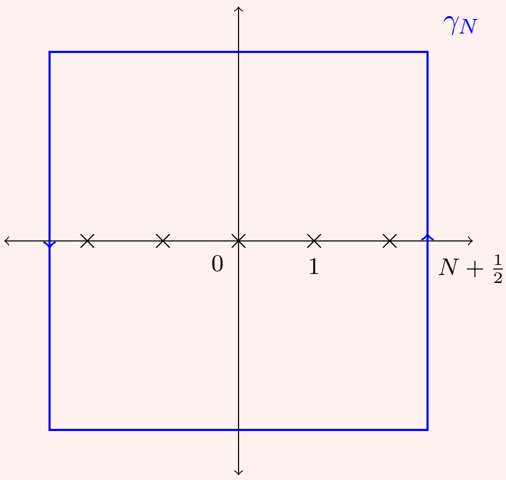
and a pole of order 3 at 0. There, from the Taylor expansions of cos and sin,

$$\begin{aligned}\cot z = \frac{1 - z^2/2 + O(z^4)}{z(1 - z^2/6 + O(z^4))} &= \frac{1}{z}\left(1 - \frac{z^2}{2} + O(z^4)\right)\left(1 + \frac{z^2}{6} + O(z^4)\right) \\ &= \frac{1}{z} - \frac{z}{3} + O(z^3),\end{aligned}$$

so that

$$f(z) = \frac{\pi \cot(\pi z)}{z^2} = \frac{1}{z^3} - \frac{\pi^2}{3z} + O(z), \quad \text{giving} \quad \operatorname{Res}_{z=0} f = -\frac{\pi^2}{3}.$$

Now, for $N \in \mathbb{N}$, let γ_N be the positively oriented boundary of the square with vertices $(\pm(N + \frac{1}{2}), \pm(N + \frac{1}{2}))$.



The square passes half-way between consecutive integers, so it encloses the poles $0, \pm 1, \ldots, \pm N$, each with winding number 1. By the residue theorem,

$$\int_{\gamma_N} f(z)\, \mathrm{d}z = 2\pi i \left(2\sum_{n=1}^{N} \frac{1}{n^2} - \frac{\pi^2}{3}\right).$$

On the other hand, one can check that $|\cot(\pi z)|$ is bounded on γ_N by a constant C independent of N (on the vertical sides $\cot(\pi z)$ is purely imaginary with modulus at most 1; on the horizontal sides $|\cot(\pi z)| \le \coth(\pi/2)$). Since $|z| \ge N + \frac{1}{2}$ on the square and $\text{length}(\gamma_N) = 4(2N+1)$,

$$\left|\int_{\gamma_N} f(z)\, \mathrm{d}z\right| \le \frac{\pi C}{(N + \frac{1}{2})^2} \cdot 4(2N+1) \to 0 \quad \text{as } N \to \infty.$$

Letting $N \to \infty$ in the previous display therefore gives

$$2\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{3}, \quad \text{that is} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The same method, with $\pi\cot(\pi z)$ replaced by $\pi\operatorname{cosec}(\pi z)$ and other kernels, evaluates a whole family of classical series.

§8 The Argument Principle and Rouché's Theorem

Our final circle of ideas uses the residue theorem to *count*. The quantity $\frac{f'}{f}$ – the 'logarithmic derivative' of f – turns out to have poles exactly at the zeros and poles of f, with residues recording their orders, and integrating it counts zeros and poles inside a contour.

§8.1 The Argument Principle

Proposition 8.1 (Logarithmic Derivatives)

If f has a zero (respectively, a pole) of order k at $z = a$, then f'/f has a simple pole at a with residue k (respectively $-k$).

Proof. If a is a zero of order k, write $f(z) = (z-a)^k g(z)$ on some disk $D(a,r)$, with g holomorphic and non-vanishing there. Then

$$\frac{f'(z)}{f(z)} = \frac{k(z-a)^{k-1}g(z) + (z-a)^k g'(z)}{(z-a)^k g(z)} = \frac{k}{z-a} + \frac{g'(z)}{g(z)},$$

and g'/g is holomorphic on $D(a,r)$, so f'/f has a simple pole at a with residue k. For a pole of order k, write $f(z) = (z-a)^{-k}g(z)$ and the same computation gives residue $-k$. □

Definition 8.2

For a zero or pole a of f, we write $\operatorname{ord}_f(a)$ for its order.

Theorem 8.3 (Argument Principle)

Let f be a meromorphic function on a domain U, with finitely many zeros $a_1, \ldots, a_k$ and poles $b_1, \ldots, b_\ell$. Let γ be a closed curve in U, homologous to zero in U, with no zero or pole of f on the curve. Then

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z = \sum_{i=1}^{k} I(\gamma; a_i)\operatorname{ord}_f(a_i) - \sum_{j=1}^{\ell} I(\gamma; b_j)\operatorname{ord}_f(b_j).$$

Proof. The function $g = f'/f$ is holomorphic away from the zeros and poles of f (at a removable singularity of f, take the holomorphic extension; if that extension is non-vanishing there, g is holomorphic there too, and if it vanishes we have simply found another zero). So the singularities of g in U are precisely $a_1, \ldots, a_k, b_1, \ldots, b_\ell$, all simple poles with residues $\operatorname{ord}_f(a_i)$ and $-\operatorname{ord}_f(b_j)$ by the proposition. Now apply the residue theorem to g. □

Why is this called the *argument* principle? Let $\Gamma = f \circ \gamma$, a closed curve avoiding 0 (since f has no zeros or poles on γ). Then, writing $[a,b]$ for the parameter interval of γ,

$$I(\Gamma;0) = \frac{1}{2\pi i}\int_\Gamma \frac{\mathrm{d}w}{w} = \frac{1}{2\pi i}\int_a^b \frac{f'(\gamma(t))\gamma'(t)}{f(\gamma(t))}\,\mathrm{d}t = \frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z.$$

So the left-hand side of the argument principle is precisely *the number of times $f\circ\gamma$ winds around the origin* – the total change in $\arg f$ along γ, divided by 2π. Zeros inside make the image wind anticlockwise; poles, clockwise.

To turn the argument principle into clean counting statements, we want curves for which all the winding numbers are 0 or 1.

Definition 8.4 (Bounding a Domain)

Let Ω be a domain and γ a closed curve. We say γ **bounds** Ω if

$$I(\gamma;w) = \begin{cases} 1 & \text{for } w\in\Omega, \\ 0 & \text{for } w\notin \Omega\cup\operatorname{image}(\gamma). \end{cases}$$

For example, $\partial D(0,1)$ bounds $D(0,1)$. If γ bounds Ω then Ω is bounded (all winding numbers vanish far away), and one can check $\partial\Omega\subseteq\operatorname{image}(\gamma)$. The fundamental supply of such curves comes from the *Jordan curve theorem*: every simple closed curve γ (one with no self-intersections) splits the plane into exactly two components, one bounded and one unbounded, and γ or $-\gamma$ bounds the bounded component. This is intuitively obvious and surprisingly hard to prove; we will take it as known, and in any concrete example the winding numbers can be checked by hand anyway.

Corollary 8.5 (Counting Zeros and Poles)

Let γ be a closed curve bounding a domain Ω, and let f be meromorphic on an open set $U\supseteq\Omega\cup\operatorname{image}(\gamma)$, with no zeros or poles on $\operatorname{image}(\gamma)$. Then f has finitely many zeros and poles in Ω, and if N and P denote the number of zeros and poles respectively, counted with multiplicity,

$$N-P = \frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,\mathrm{d}z = I(f\circ\gamma;0).$$

Proof. First, finiteness. The closure $\overline{\Omega}$ is compact (as Ω is bounded) and contained in $\Omega\cup\operatorname{image}(\gamma)\subseteq U$. If f had infinitely many poles in $\overline{\Omega}$, they would accumulate at some $w\in\overline{\Omega}\subseteq U$; but the singular set of a meromorphic function is discrete, and a non-singular point has a neighbourhood free of singularities – either way, a contradiction. If f had infinitely many zeros in $\overline{\Omega}$, they would accumulate at some $w\in\overline{\Omega}$, which is either a regular point or a removable singularity of f (it cannot be a pole, since $f\to\infty$ at a pole but $f=0$ along the accumulating zeros); then by the identity theorem $f\equiv 0$ near w, and hence (applying the identity theorem on the domain Ω minus the singular set, which is still connected) f vanishes throughout Ω, contradicting $f\neq 0$ on the boundary curve.

Since γ bounds Ω, we have $I(\gamma;w)=0$ for every $w\notin\Omega\cup\operatorname{image}(\gamma)$; in particular γ is homologous to zero in U. Every zero or pole in Ω has winding number 1, and those outside contribute nothing, so the argument principle gives exactly $N-P$;

and we identified the integral with $I(f \circ \gamma; 0)$ above. □

§8.2 The Local Degree and the Open Mapping Theorem

The argument principle also tells us what a holomorphic map looks like *locally*.

Definition 8.6 (Local Degree)

Let f be holomorphic and non-constant on a disk $D(a, R)$. The **local degree** of f at a, written $\deg_f(a)$, is the order of the zero of $f(z) - f(a)$ at $z = a$.

Note this is a well-defined positive integer: $f(z) - f(a)$ is not identically zero (else f is constant), so has a zero of some finite order $d \geq 1$ at a. For instance, $f(z) = (z-1)^4 + 1$ has $\deg_f(1) = 4$.

Theorem 8.7 (Local Degree Theorem)

Let $f : D(a, R) \to \mathbb{C}$ be holomorphic and non-constant with $\deg_f(a) = d$. Then there is $r_0 > 0$ such that for every $r \in (0, r_0]$ there is $\varepsilon > 0$ with the property: for every w with $0 < |w - f(a)| < \varepsilon$, the equation $f(z) = w$ has exactly d distinct roots in $D(a, r) \backslash \{a\}$.

Proof. Let $g(z) = f(z) - f(a)$. Since g is non-constant, $g \not\equiv 0$ and $g' \not\equiv 0$ on $D(a, R)$, so by the principle of isolated zeros we may choose $r_0 \in (0, R)$ such that g and g' have no zeros on $\overline{D(a, r_0)} \backslash \{a\}$.

Fix $r \in (0, r_0]$, let $\gamma(t) = a + re^{2\pi i t}$, and let $\Gamma = g \circ \gamma$. The image of Γ is compact and misses 0 (as $g \neq 0$ on the circle), so there is $\varepsilon > 0$ with $D(0, \varepsilon) \cap \mathrm{image}(\Gamma) = \emptyset$.

Now let $0 < |w - f(a)| < \varepsilon$. Since winding numbers are constant on the connected set $D(0, \varepsilon)$,

$$I(\Gamma; w - f(a)) = I(\Gamma; 0).$$

By our counting corollary applied to g on the disk (the circle γ bounds $D(a, r)$), $I(\Gamma; 0)$ is the number of zeros of g in $D(a, r)$, which is d (the single zero at a, counted with multiplicity d). On the other hand, $I(\Gamma; w - f(a))$ counts the zeros of $g(z) - (w - f(a)) = f(z) - w$ in $D(a, r)$, with multiplicity. So $f(z) = w$ has d roots in $D(a, r)$ counted with multiplicity.

Finally, these roots are distinct and none equals a: since $w \neq f(a)$, $z = a$ is not a root; and at any root $z \neq a$ we have $f'(z) = g'(z) \neq 0$ by our choice of r_0, so each root is simple. □

So near a point of local degree d, a holomorphic map is d-to-one onto a neighbourhood of the image point (think of $z \mapsto z^d$ near 0, which is the model case). One consequence deserves to be a headline theorem.

Theorem 8.8 (Open Mapping Theorem)

A non-constant holomorphic function on a domain maps open sets to open sets.

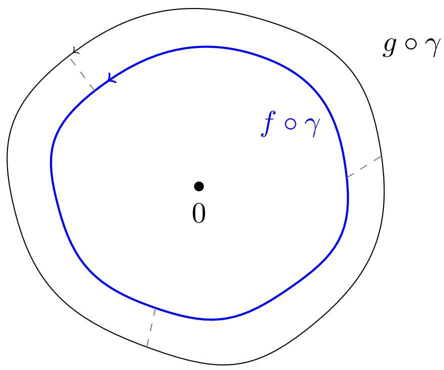
Proof. Let $f : U \to \mathbb{C}$ be holomorphic and non-constant on the domain U, let $V \subseteq U$ be open, and let $b = f(a) \in f(V)$ with $a \in V$. Note f is non-constant on every disk in U (by the identity theorem, constancy on a disk forces constancy on U). Choose $r > 0$ small enough that $D(a, r) \subseteq V$ and the local degree theorem applies with some $\varepsilon > 0$; then every $w \in D(b, \varepsilon)$ with $w \neq b$ is attained by f on $D(a, r)$, and b itself is attained at a. Hence $D(b, \varepsilon) \subseteq f(D(a, r)) \subseteq f(V)$, so $f(V)$ is open. □

Remark. Compare with the real situation: $x \mapsto x^2$ maps the open interval $(-1, 1)$ onto $[0, 1)$, which is not open. Once again holomorphic functions are far more rigid than smooth real ones.

The local degree theorem also settles a promise from Section 1: *if f is holomorphic and injective, then $f' \neq 0$ everywhere.* Indeed, if $f'(a) = 0$ then $f(z) - f(a)$ has a zero of order $d \geq 2$ at a, and the local degree theorem produces points near a where f takes the same value $d \geq 2$ times – contradicting injectivity. So in our definition of conformal equivalence, the non-vanishing derivative condition was automatic.

§8.3 Rouché's Theorem

Our last theorem is a wonderfully practical tool for locating zeros. The idea is often called 'walking the dog'. Suppose you walk a closed loop around a lamppost (the origin), holding a dog on a lead. If the lead always stays shorter than your distance to the lamppost, then the dog is dragged around the lamppost exactly as many times as you walk around it – the dog can never sneak an extra loop, nor miss one. Here you are $g \circ \gamma$, the dog is $f \circ \gamma$, and the lead has length $|f - g|$.



Theorem 8.9 (Rouché's Theorem)

Let γ be a closed curve bounding a domain Ω, and let f, g be holomorphic on an open set $U \supseteq \Omega \cup \text{image}(\gamma)$. If

$$|f(z) - g(z)| < |g(z)| \quad \text{for all } z \in \text{image}(\gamma),$$

then f and g have the same number of zeros in Ω, counted with multiplicity.

Proof. The strict inequality forces $g \neq 0$ on the curve (else $0 \leq |f - g| < 0$) and then also $f \neq 0$ on the curve ($|f| \geq |g| - |f - g| > 0$). Let $h = f/g$, which is meromorphic on U, holomorphic and non-vanishing on an open set containing image(γ). Both f and g have finitely many zeros in Ω by our counting corollary (neither vanishes identically, as neither vanishes on the curve).

The hypothesis says exactly that

$$|h(z) - 1| < 1 \quad \text{on } \operatorname{image}(\gamma),$$

so the closed curve $h \circ \gamma$ lies in the disk $D(1, 1)$, which does not contain 0. Since the winding number about 0 vanishes on the unbounded component of the complement of $h \circ \gamma$, and 0 can be joined to infinity without crossing $D(1, 1)$, we get $I(h \circ \gamma; 0) = 0$.

By the argument principle (counting corollary) applied to h,

$$0 = I(h \circ \gamma; 0) = N_h - P_h,$$

where N_h, P_h are the numbers of zeros and poles of h in Ω with multiplicity. Now at any point $w \in \Omega$, writing ord_w for the order of vanishing at w (negative at a pole), we have $\operatorname{ord}_w(h) = \operatorname{ord}_w(f) - \operatorname{ord}_w(g)$, and summing over all relevant points,

$$N_h - P_h = \sum_{w \in \Omega} \operatorname{ord}_w(h) = \sum_{w \in \Omega} \operatorname{ord}_w(f) - \sum_{w \in \Omega} \operatorname{ord}_w(g) = N_f - N_g,$$

where N_f, N_g count the zeros of f and g in Ω with multiplicity (the sums are finite, over the zeros of f and g). Hence $N_f = N_g$. □

Example 8.10 (Locating Roots of $z^4 + 6z + 3$)

Let $f(z) = z^4 + 6z + 3$. We claim f has exactly three roots (with multiplicity) in the annulus $\{1 < |z| < 2\}$.

On the circle $|z| = 2$: taking $g(z) = z^4$, we have

$$|f(z) - g(z)| = |6z + 3| \leq 15 < 16 = |z^4| = |g(z)|,$$

so by Rouché's theorem f has the same number of zeros in $D(0, 2)$ as z^4, namely 4. So all four roots of f lie in $|z| < 2$ (and none on $|z| = 2$, by the strict inequality).

On the circle $|z| = 1$: taking $g(z) = 6z$ instead,

$$|f(z) - g(z)| = |z^4 + 3| \leq 4 < 6 = |6z| = |g(z)|,$$

so f has as many zeros in $D(0, 1)$ as $6z$, namely one (and none on $|z| = 1$).

Subtracting, exactly $4 - 1 = 3$ roots lie in $\{1 < |z| < 2\}$.

Example 8.11 (A Perturbation Result)

Rouché's theorem explains why roots of polynomials move continuously with the coefficients. For instance, suppose $p(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_0$ and z_0 is a simple root. For any sufficiently small $\delta > 0$, we have $m = \min_{|z - z_0| = \delta} |p(z)| > 0$, and then any polynomial q with $|q(z) - p(z)| < m$ on the circle $|z - z_0| = \delta$ has, by Rouché's theorem (with $g = p$, $f = q$), exactly one root in $D(z_0, \delta)$. Sufficiently small changes of the coefficients therefore move the root only slightly.

And that – from a single innocuous-looking limit defining $f'(w)$, all the way to counting the roots of polynomials by walking dogs around lampposts – is the IB Complex Analysis course.